

Orthopedic Implant–Associated Infections

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Short View Summary

PERIPROSTHETIC JOINT INFECTION (PJI)

Diagnosis

- Determination that a prosthetic joint is infected depends on recovery of an organism from a joint aspirate or surgically obtained material from the joint. Synovial or tissue inflammation, wound dehiscence, and a draining sinus tract are also confirmatory signs of PJI. Joint effusion and early loosening of the prosthesis can point toward this diagnosis.

Microbiology

- The most frequent microorganisms are *Staphylococcus aureus*, coagulase-negative staphylococci, and streptococci.
- *Cutibacterium acnes* is the most important microorganism in periprosthetic shoulder infection.
- Microorganisms establish a biofilm on the surface of the implant.

Pathogenesis

- Foreign devices are covered by host proteins (e.g., fibronectin) after implantation, favoring bacterial adherence.
- The minimal infecting dose is greater than 10,000-fold lower in the presence than in the absence of an implant.
- Granulocyte function around the implant is impaired (activation and degranulation).

Clinical Manifestations

- Acute exogenous PJI produces local signs of inflammation (wound dehiscence, secretion, erythema).
- Acute hematogenous PJI involves new-onset pain at any time after implantation, initially without local signs of infection, and new-onset articular effusion.
- In chronic PJI, pain occurs because of early loosening, local inflammation, sinus tract, and chronic articular effusion.

Treatment

- The cornerstone of successful treatment is early diagnosis.
- Cure is possible only with adequate surgery combined with appropriate antibiotic therapy.
- A treatment algorithm (see [Fig. 107.1](#)) allows one to choose the most appropriate surgical intervention: débridement with retention, one-stage exchange, two-stage exchange, removal without replacement, or suppressive therapy.
- Guidelines are available at

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INTERNAL FIXATION–ASSOCIATED INFECTION

Diagnosis

INTERNAL FIXATION–ASSOCIATED INFECTION

Diagnosis

- Diagnosis of infection associated with implanted devices used to stabilize bone fractures, such as pins and rods, requires culture of the device, usually after surgical removal. Infection related to fracture fixation is defined as fracture-related infection (FRI). The definition uses suggestive and confirmatory criteria, based on clinical, microbiologic, and histopathologic findings.

Microbiology

- Microbiology is comparable with that of PJI. Staphylococci are the most important pathogens.
- In open fractures, the wound is contaminated with environmental bacteria.

Pathogenesis

- The exogenous route plays a more important role than the hematogenous route.
- Infection may also occur via an adjacent focus (contiguous).

Clinical Manifestations

- Acute early postoperative infections cause wound healing disturbances, discharge, and erythema.
- In the case of acute symptoms that occur after an uneventful postoperative period, if systemic infection signs are dominating, pain is the most important local sign. Other local features are absent in the beginning and become apparent in the course of disease.
- Acute symptoms may also be caused by reactivation of chronic posttraumatic osteomyelitis that has been silent for many years. Pain may occur without prominent systemic inflammatory signs.
- Chronic symptoms include sinus tract, pain, and implant loosening.

Treatment

- Bone fractures are less susceptible to infection if stabilized.
- In acute infections, retain implant with débridement and antimicrobial therapy until the fracture is consolidated.
- In chronic infections, the implant must be removed. When delayed infections on implants are suppressed with antimicrobials, the hardware must be removed after healing of bone fracture.

Orthopedic devices are used to stabilize bone fractures, fuse the vertebral column, correct deformities such as scoliosis, and replace damaged joints. ¹⁻³ Whereas internal stabilization devices could be removed after a bone fracture is healed, prosthetic joints are kept in the body as long as their function remains full and pain free. The presence of an implant increases the risk for an infection because its local susceptibility for bacterial adherence is high. ⁴ Thus, even with the most meticulous infection control, implant-associated infections occur. The optimal management of orthopedic implant-associated infections requires close cooperation of experienced specialists in infectious diseases, microbiology, orthopedic and trauma surgery, and plastic surgery. ^{5,6}

Periprosthetic Joint Infection

Definition

The classic Centers for Disease Control and Prevention definition of wound infection is not ideal for implant-associated infections, ⁷ because superficial and deep wound infection cannot reliably be differentiated clinically. ⁸ Therefore, even in the case of a presumably “superficial” wound infection, revision surgery must be considered. Empirical antibiotic treatment without diagnostic workup should be avoided in the absence of life-threatening sepsis.

The definition of periprosthetic joint infection (PJI) should have a high sensitivity not to delay the diagnosis of PJI. For clinical studies, different definitions have been proposed by the International Consensus Meeting on Orthopedic Infections

clinical studies, different definitions have been proposed by the International Consensus Meeting on Orthopedic Infections, the Musculoskeletal Infection Society, the Infectious Diseases Society of America, and the European Bone and Joint Infection Society (EBJIS).^{3, 9-12} The main problem with these definitions is the risk of underdiagnosing low-grade infection. Therefore, the EBJIS developed a nonbinary definition.¹⁰ Three levels of probability (infection unlikely, infection likely, infection confirmed) were proposed, allowing clinical decision-making. **Table 107.1** shows the criteria for the confirmed infection. C-reactive protein (CRP) is not a sensitive parameter in patients with PJI caused by low-virulence microorganisms.¹³ The detection of microorganisms in intraoperative specimens from synovial fluid, tissue, or bone cement fragments is important. Swabs from the operative incision site or from [Page 1367](#) a draining sinus are difficult to interpret and are not recommended. As a criterion for PJI, detection of the same microorganisms in at least two samples is required. A single positive culture of a pathogen belonging to the skin microbiome (e.g., *Cutibacterium acnes*, coagulase-negative staphylococci) does not confirm PJI.

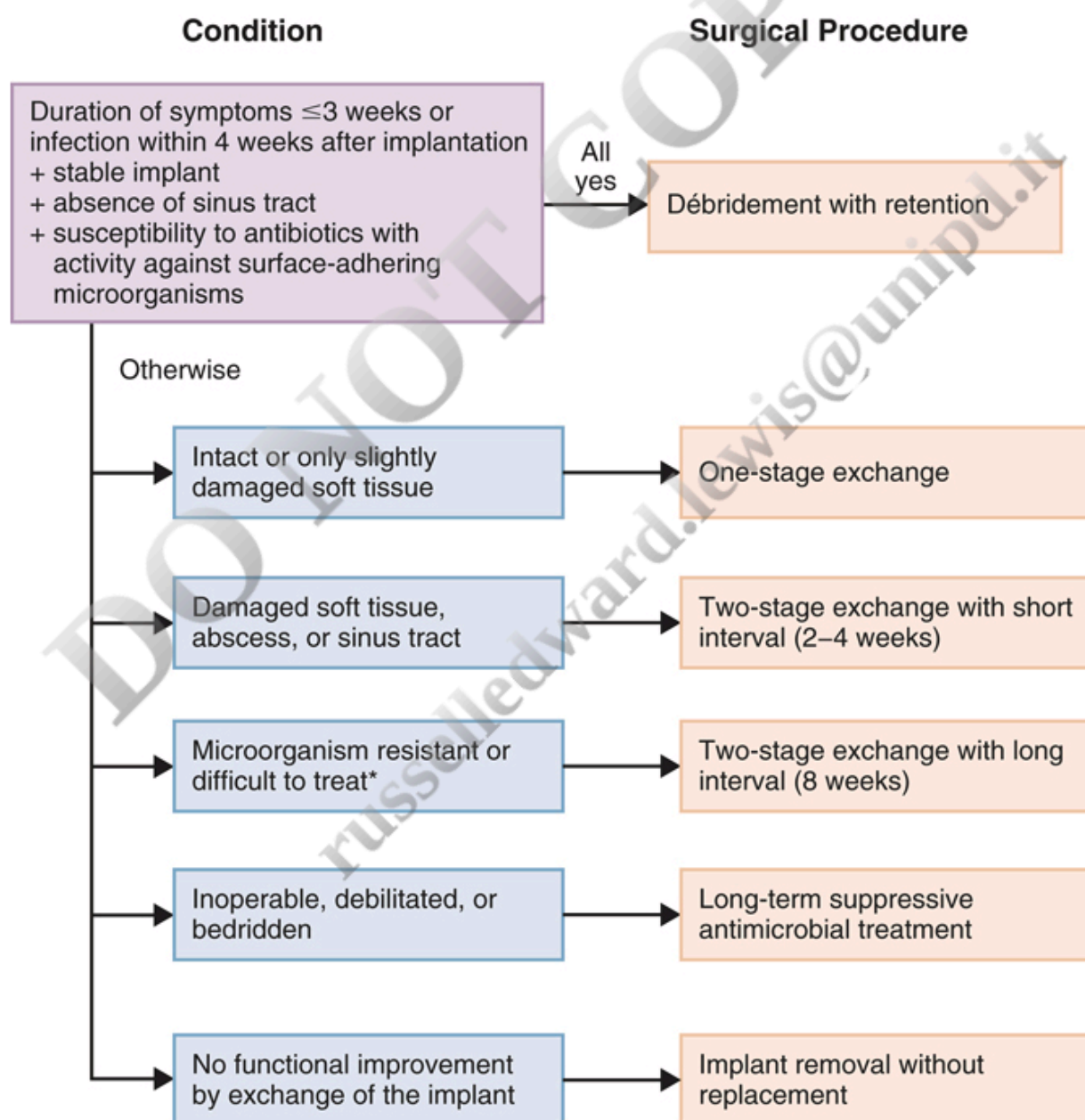


FIG. 107.1 Surgical treatment algorithm for prosthetic joint infections.*Difficult-to-treat microorganisms include microorganisms resistant to antibiotics with good oral bioavailability,

FIG. 107.1 Surgical treatment algorithm for prosthetic joint infections.*Difficult-to-treat microorganisms include microorganisms resistant to antibiotics with good oral bioavailability, rifampin-resistant staphylococci, quinolone-resistant gram-negative bacilli, and fungi. *Modified from Trampuz A, Zimmerli W. Prosthetic joint infections: update in diagnosis and treatment. Swiss Med Wkly. 2005;135:243–251.*

[Follow for extended description](#)

TABLE 107.1

Definition of Confirmed Periprosthetic Joint Infection According to the EBJIS Definition¹⁰ (any positive finding)

- Clinical features: Sinus tract with evidence of communication to the joint or visualization of the prosthesis
- Synovial fluid cytological analysis: >3000 leukocytes/ μ L or >80% neutrophils
- Synovial fluid biomarkers: Positive α -defensin (not valid in cases of adverse local tissue reaction, hematomas, or acute inflammatory arthritis, including gout)
- Microbiology: ≥ 2 positive intraoperative fluid or tissue samples with the same microorganism. If antibiotic treatment has been given (not simple prophylaxis), the results may be compromised. In these cases, molecular techniques may have a place.
- Microbiology: >50 CFU/mL of any organism in sonication fluid
- Histology: Presence of ≥ 5 neutrophils in >5 HPF (>400 $\times$ magnification) or presence of visible microorganisms

CFU, Colony-forming units; EBJIS, European Bone and Joint Infection Society; HPF, high-power field.

Pathogenesis

Prosthetic joints are made of metal and polymers, most commonly polyethylene inlay between metal shell and metal head. Bone cement (i.e., polymethyl methacrylate [PMMA]) is frequently employed at the metal-bone interface. Biocompatible materials do not cause inflammation in the absence of infection and are therefore selected for implantation.¹⁴

After implantation, the device is covered by host proteins (e.g., fibronectin) favoring bacterial adherence.¹⁵ Complement in interstitial fluid can be activated at the surface of polymeric implants, such as polyethylene particles, which results in degranulation of local neutrophils.⁴ In addition, it has been shown that granulocytes interact with the nonphagocytosable surface of implants. This process is called *frustrated phagocytosis* and results in impaired granulocyte function.^{4, 16} Implants and polymer particles, which are produced in variable amounts after arthroplasty, compromise granulocyte function.¹⁷ This interaction with wear particles, fragments from within the joint space, results in impaired granulocyte function and can lead to aseptic loosening of joint prostheses.¹⁸

Implanted foreign material is highly susceptible to local infection. In 1957, Elek and Conen¹⁹ showed in human volunteers that in the vicinity of suture material, the minimal abscess-forming dose is as low as 100 colony-forming units of *Staphylococcus aureus*, which is more than 10,000-fold lower than in the absence of foreign material. This observation was reproduced in animal models of foreign body–associated infection using different gram-positive and gram-negative microorganisms.^{17, 20–23} The route of infection can be exogenous, hematogenous, or contiguous spread. The last route is rare and implicates spread from an adjacent focus of infection (e.g., osteomyelitis, deep soft tissue infection). A guinea pig

microorganisms. The route of infection can be exogenous, hematogenous, or contiguous spread. The last route is rare and implicates spread from an adjacent focus of infection (e.g., osteomyelitis, deep soft tissue infection). A guinea pig model of *S. aureus* bacteremia demonstrated that microorganisms can seed on an implant from a distant site by the hematogenous route.²⁴ This parallels observations in clinical practice.²⁵⁻²⁷ Finally, the exogenous route includes a joint inoculation of microorganisms during surgery or in the early postoperative period. Exogenous infections can also occur later via a penetrating event (e.g., trauma, injection).

Microorganisms adhering to implants within a biofilm are protected from phagocytosis.²⁸ In addition, most antibiotics do not kill adherent bacteria despite penetration into the biofilm.^{29,30} The lack of efficacy of many antimicrobial agents depends on the low growth rate of organisms in the biofilm.³¹ As an exception, rifampin can eliminate biofilm-associated gram-positive microorganisms,^{20,32-36} and fluoroquinolones are active against biofilms from gram-negative bacilli, provided that the biofilm is acute.^{22,23,37}

Classification

No classification system of PJI is unanimously accepted. On the basis of the two major infection routes, PJI can be classified as either exogenous or hematogenous infection.³⁸ In considering the virulence of bacteria and the host response of immunocompetent patients, PJI can be classified on the basis of symptoms as acute or chronic. The traditional classification differentiates among early (those developing within <3 months after implantation), delayed (3–24 months after surgery), and late (>24 months after implantation) infection.^{3,12} In clinical practice, it is more useful to classify PJI as follows:

- Early postinterventional PJI: Infection that manifests within 1 month after an invasive procedure (surgery or arthrocentesis)
- Acute hematogenous PJI: Infection with 3 weeks' duration of symptoms or less after an uneventful postoperative period
- Chronic PJI: Infection with symptoms that persist for more than 3 weeks and are beyond the early postinterventional period

This simple classification allows the differentiation between acute PJI, including both early postinterventional and acute hematogenous PJI, which can be potentially cured with débridement and implant retention (DAIR), and chronic PJI which requires device removal for cure.^{3,12} Acute hematogenous infection is generally caused by virulent bacteria, and débridement is required in order for the device to be retained. This type of infection can occur at any time after surgery, as long as the implant is in the body.^{25-27,39,40} Hematogenous PJI may be misclassified as “acute” if it is caused by reactivation from a persisting silent PJI caused by an earlier bacteremia. Careful exploration of the case history may reveal this misclassification. Early postoperative PJI manifests with poor wound healing, joint fluid accumulation, or persisting pain. In these cases, early detection with a high degree of suspicion, rapid diagnostic workup, and prompt surgical treatment are required for retention of the implant. If these (exogenous) infections are not detected early after surgery (within the first month) or if symptoms persist for more than 3 weeks, or both, chronic PJI develops, irrespective of the infection route. This form of infection requires removal of all hardware and bone cement for cure.

Risk Factors

Several risk factors have been described for the development of PJI.^{5,8,41-43} Most data are derived from patients with total hip and knee arthroplasties. Some of the proposed risk factors have to be interpreted with caution because the corresponding studies used different statistical methods or focused on only one particular anatomic site. Risk factors for acquiring PJI can be categorized as patient characteristics, surgery-related and postoperative factors, and risk during bacteremia.^{42,44} Gahr and colleagues⁴⁵ showed that the individual risk for PJI can be reliably predicted knowing the

bacteremia. ^{43, 44} Goltz and colleagues ⁴⁵ showed that the individual risk for PJI can be reliably predicted by using the American College of Surgeons (ACS) National Surgical Quality Improvement Program (NSQIP) Surgical Risk Calculator (<http://riskcalculator.facs.org/>).

Patient Characteristics

Diabetes mellitus, obesity (body mass index ≥ 30 weight in kg/height in m²), rheumatic diseases, and immunosuppressive therapy are among the most frequently reported risk factors for developing PJI. These factors influence wound healing and predispose to many infectious diseases. Congestive heart failure, chronic pulmonary disease, peripheral vascular disease, malignancy, coagulopathy, and preoperative anemia, among other comorbidities, have been described as risk factors for PJI. Preoperative asymptomatic bacteriuria is not a risk factor. ^{46, 47} Overall, it is rather the “net state of polymorbidity” of a patient (e.g., number of comorbidities, physical status) than a single variable that plays a unique role in infection pathogenesis. Finally, the patient’s history with respect to previous tissue damage should be considered. Previous joint surgery, mainly previous arthroplasty, has also been identified as a risk factor.

Surgery-Related Risk Factors

High complexity surgical procedures, prolonged operation times, and multiple simultaneous joint revisions are the main surgical risk factors for PJI.

Postoperative Factors

In essence, all wound complications (e.g., delayed healing, drainage, persistent dehiscence, hematoma, seroma, wet wound after initially being dry) increase the risk for infection. Although insulin-dependent diabetes mellitus is a risk factor for postoperative hyperglycemia as compared to noninsulin dependent diabetes, it is controversial whether postoperative hyperglycemia is a risk factor for PJI. ^{42, 48} Risk factors for PJI that are not primarily associated with the surgical procedure or wound healing include atrial fibrillation, myocardial infarction, and a prolonged hospital stay.

Risk During Bacteremia

In hematogenous PJI, the cardiovascular system (endocarditis, catheter-related infection) followed by skin and soft tissue infection are the leading sites of primary infection. ⁴⁹ The predominant pathogens are *S. aureus* and *Streptococcus* spp. ⁴⁹ Indeed, the rate of PJI after *S. aureus* bacteremia is approximately 35%. ^{25- 27} This high infection rate indicates that patients with previously uninfected prosthetic joints and *S. aureus* bacteremia should be carefully monitored clinically for the development of PJI. Again, rapid diagnosis of PJI may avoid an exchange of the prosthesis because these infections can be cured with DAIR.

Microbiology

As a rule, all microorganisms can cause PJI, including mycobacteria. ^{50- 52} However, staphylococci are the most commonly isolated microorganisms. In six studies with a total of 2578 episodes of PJI, the following microorganisms were detected. ^{8, 53- 57}

- *S. aureus*, 21% to 43%
- Coagulase-negative staphylococci, 17% to 39%
- Streptococci, 7% to 12%
- Gram-negative aerobic bacilli, 5% to 12%
- Enterococci, 1% to 8%
- Anaerobic bacteria, 2% to 8%

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Other microorganisms such as diphtheroids and *Candida* spp. cause PJI less frequently. *Corynebacterium acnes* is responsible for about 3% of periprosthetic hip and knee infection but up to 38% of periprosthetic shoulder infection. 56· 58 In about 15%, polymicrobial infection occurs. 59 Infections with mixed flora are mainly observed in patients with protracted wound healing, resulting in exogenous superinfection. 60 In 4% to 15% of cases, no microorganism is detected. 8· 53-57· 59· 61 This proportion depends on the frequency of pretreatment with antimicrobial agents, type of microorganism, and microbiologic technique.

Clinical Manifestations and Differential Diagnosis

Acute Periprosthetic Joint Infection

The description of the clinical manifestation stems mainly from hip and knee PJI. However, PJI in other joints may manifest differently. 62 Acute exogenous infection typically causes local signs of inflammation, whereas acute hematogenous infection is characterized by new-onset pain, initially without prominent local signs of infection. Acute infections are usually caused by virulent pathogens, mainly *S. aureus*, β -hemolytic streptococci, and less frequently gram-negative bacilli. 3· 63

Most early (postoperative) acute infections have an exogenous origin. Wound dehiscence, drainage, and erythema are typical signs. A systemic inflammatory response syndrome may be missing. 3· 38 The differential diagnosis of postoperative acute PJI includes wound complications such as hematoma or seroma. Because the clinical differentiation of superficial and deep wound infection is not reliable, 8 each suspicious wound needs a careful orthopedic evaluation (e.g., surgical exploration of the site) in order to establish or exclude PJI. 64 Consideration of this principle increases the fraction of patients who can be cured with DAIR. Page 1369

Acute-onset PJI occurring beyond the postoperative period is generally of hematogenous origin (acute hematogenous PJI). In such infections, systemic signs of inflammation are prominent. The patient typically reports profound pain at the site of the implant, although soft tissue damage is less frequent and less obviously visible than in early exogenous PJI. 3· 38 The differential diagnosis of acute PJI after an uneventful postsurgical period is narrow. It includes crystal arthropathy, which can be detected with microscopic examination of the synovial fluid and biopsy samples. 65 Venous thrombosis and arterial emboli in vessels that are in close anatomic proximity to the joint can also imitate clinical signs of PJI.

All patients with acute symptoms, irrespective of the interval between prosthesis implantation and clinical manifestation, require rapid diagnostic workup because the implant can potentially be retained if symptom duration is short. 3· 12· 64 Seemingly acute hematogenous PJI may be exacerbated latent PJI. Therefore, case history since implantation should be carefully taken, in order to choose the correct surgical procedure. 66· 67

Chronic Periprosthetic Joint Infection

Chronic PJI can be exogenous or hematogenous. If the joint is infected at surgery by low-virulence organisms, infection often manifests beyond the early postoperative period (≥ 1 month). When symptoms persist for several weeks, irrespective of infection route, the PJI becomes chronic. The key symptoms are chronic joint effusion, pain caused by local inflammation or implant loosening, and, occasionally, sinus tracts. In addition, CRP and erythrocyte sedimentation rate (ESR) do not normalize after surgery and typically fluctuate within a slightly elevated range. However, these tests are not specific. The differential diagnosis includes mechanical failure, excessive wear debris, or metal-induced delayed type

hypersensitivity. 68· 69 The sensitivity of CRP is limited in patients with low-virulence microorganisms. It has been shown to be normal (< 8 mg/L) in 32% of patients with documented chronic PJI. 13 If early loosening is caused by mechanical failure, one-stage exchange is appropriate, whereas in patients with septic loosening the surgical strategy depends on the

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failure, one-stage exchange is appropriate, whereas in patients with septic loosening the surgical strategy depends on the infecting agent. ⁷⁰ Thus, it is crucial to identify PJI cases before definitive surgery. The consequence of missing a low-grade infection is an inadequate revision arthroplasty and subsequent failure. In view of the generally long-lasting course in chronic PJI, there is sufficient time for thorough diagnostic procedures.

Diagnostic Procedures

Blood Tests

Leukocyte counts are diagnostically not helpful owing to poor sensitivity and low predictive values. ⁷¹ CRP and ESR are routinely used in the diagnostic workup of PJI. In a systematic review including 30 studies with various cutoffs for ESR and CRP, the sensitivities are 75% and 88%, respectively, and the specificities 70% and 74%, respectively. ⁷¹ Sensitivity is much less for low-virulence organisms, such as *C. acnes*. ^{13, 72, 73} Serum D-dimer has been proposed as minor criterion for the diagnosis of PJI. ¹¹ However, D-dimer serum levels cannot distinguish between PJI and aseptic loosening. ^{64, 74} Procalcitonin (PCT), which is a promising biomarker in patients with respiratory tract infections, has not proved helpful in differentiating septic from aseptic loosening. ^{75, 76} In addition, postoperative serum PCT levels are rarely elevated in patients with PJI without sepsis syndrome. ⁷⁷ In contrast, interleukin-6 (IL-6) values above the cutoff value of 10 pg/L appear to have an excellent sensitivity of nearly 100% to predict the presence of PJI. ^{71, 75}

Synovial Fluid Cell Counts

The threshold of leukocyte counts in synovial fluid for the diagnosis of PJI is much lower than for septic native joint arthritis. Published cutoff values show a sensitivity and specificity of about 90%. Because of differences in counting methods (automated vs. manual) and interlaboratory variations in counting results, there is no uniformly accepted precise cutoff value. In hip arthroplasty, Schinsky and colleagues ⁷⁸ reported optimal cutoff values of more than 4200 leukocytes per microliter, neutrophil fraction greater than 80%, or both. In patients with periprosthetic knee infection, Trampuz and colleagues ⁷⁹ defined an optimal cutoff value of more than 1700 leukocytes per microliter, neutrophil fraction greater than 65%, or both. Similar results were shown by Ghanem and colleagues ⁸⁰ for periprosthetic knee infection (more than 1100 leukocytes per microliter, >64% neutrophil fraction, or both). In these studies, patients with rheumatoid arthritis or joint hemorrhage, or those in the early postoperative period, were excluded. However, similar cutoff values were reported in synovial fluid of patients with and without inflammatory arthritis in periprosthetic hip infection, namely 3450 leukocytes per microliter versus 3444 leukocytes per microliter, and 78% and 75% neutrophil fraction, respectively. ⁸¹ In synovial fluid drawn shortly after surgery, the proposed cutoff values are much higher, namely, more than 25,000 leukocytes per microliter. ⁸² In summary, leukocyte and differential counts are key parameters for the diagnosis of PJI, especially in patients with negative culture.

Synovial Fluid α -Defensin

α -Defensin is an antimicrobial peptide that is released by neutrophils and natural killer cells in response to inflammation. The assay has been reported to assist in the diagnosis of PJI, although its usefulness is controversial. ^{83, 84} The result is not influenced by prior antibiotics, but metallosis (metal accumulation in tissue, such as from metal on metal friction) is a potential source of false-positive results. ^{85, 86} In two recent studies, a lateral flow assay for rapid detection of α -defensin in synovial fluid had a limited sensitivity of 67% and 69%, respectively. ^{87, 88} In addition, it has been shown that the sensitivity of synovial leukocyte count is higher than that of the qualitative α -defensin test (86% vs. 54%; $P < .001$). ⁸⁹ A strip test for detecting leukocyte esterase has also been used intraoperatively but has low sensitivity and gives false-positive results in bloody specimens. ⁹⁰

Synovial fluid D-lactate is a promising novel test. D-lactate is a pathogen-specific metabolite. Therefore, the specificity can be suspected to be higher than the one of α -defensin. Indeed, this biomarker had a sensitivity of 92%, and a high specificity of 89% in a cohort of 224 patients. ⁹¹ Further studies will show whether this rapid test can be used as a reliable point-of-care screening test.

Synovial Fluid Culture

Conventional microbiologic cultures of synovial fluid have a moderate sensitivity of approximately 85% but an excellent specificity of greater than 95%. ⁹² However, published results often do not stratify whether the fluid was obtained by puncture, arthroscopy, or open surgery. A negative culture result does not exclude PJI. The sample sensitivity can be improved by use of polymerase chain reaction (PCR) ⁹³ or by culturing synovial fluid in blood culture flasks. ⁹⁴

Intraoperative Samples for Culture and Histopathology

Swab cultures clearly have a lower sensitivity than cultures from periprosthetic tissue and synovial fluid, and must therefore not be used. ⁹⁴ Culturing synovial fluid and periprosthetic tissue has the best accuracy, specificity, and sensitivity. ⁹² For the diagnosis of PJI, at least three specimens should be obtained. ^{12, 95} The greatest accuracy is observed with four specimens processed with conventional culture and with three specimens cultured in blood culture bottles. ⁹⁶⁻⁹⁸

Biopsy culture of *C. acnes* requires anaerobic conditions. Incubation time depends on multiple factors (e.g., time from sampling to laboratory, material preparation, culture media, incubation techniques), and hence may vary among institutions. Recent investigations have reported an incubation time of up to 10 days if cultured under optimal anaerobic conditions. ^{99, 100}

Histopathologic specimens are difficult to interpret because the threshold of neutrophils per high-power field (HPF) varies among different experts, ranging from an average of 1 or more to 10 or more neutrophils when counting 10 HPFs. Many experts accept 5 or more neutrophils per HPF in $\times 40$ magnification as a positive indicator for PJI. ¹⁰¹ Frozen sections can be used only in centers with experienced pathologists who are able to differentiate between mechanical failure and infection. This technique informs the surgeon in the operating room whether infection can be excluded, and therefore whether the planned one-stage exchange for early loosening can be performed. ¹⁰² Page 1370 In cases of chronic PJI, we recommend paraffin sections, which have been reported to be superior to frozen sections. If possible, each tissue sample should be cut into two pieces, labeled with their precise origin, and submitted one each to microbiologic and histopathologic examination. If dividing tissue sample is not possible (e.g., bone biopsies), two samples from the same location should be obtained. The comparison between culture results and histopathologic findings helps to differentiate between contamination and infection.

Sonication

Sonication of foreign-body material, such as an implant or parts of it, followed by culturing or use of molecular methods on the sonicate fluid, or both, allows detection of microorganisms in biofilms. ¹⁰³ In a population with either PJI or aseptic failure, Trampuz and colleagues ¹⁰⁴ performed a prospective trial comparing culture results from synovial fluid, periprosthetic tissue, and sonicate fluid. In this trial, the sensitivities for detecting PJI with the three different methods were 56.3%, 60.8%, and 78.5%, respectively, with specificities greater than 98%. According to a meta-analysis of 13 studies, the sensitivity of sonication culture is significantly better than that of traditional culture. ^{105, 106} Improved sensitivity has also been reported when sonication of synovial fluid preceded PCR. ¹⁰⁷

Molecular Diagnostics

Molecular methods have been used in synovial fluid, periprosthetic tissue specimens, and sonicate fluid from explanted prostheses or modular parts of them. Different techniques have been used, such as broad-range, targeted or multiplex PCR, as well as next-generation sequencing. ^{105, 108, 109} In patients pretreated with antibiotics, the sensitivity of broad-spectrum

Molecular methods have been used in synovial fluid, periprosthetic tissue specimens, and sonicate fluid from explanted prostheses or modular parts of them. Different techniques have been used, such as broad-range, targeted or multiplex PCR, as well as next-generation sequencing. ¹⁰⁵ ¹⁰⁸ ¹⁰⁹ In patients pretreated with antibiotics, the sensitivity of broad-spectrum PCR is higher than that of culture, but still only about 70%. ⁹³ In addition, it does not give information about the susceptibility of the microorganism with a few exceptions (e.g., *mecA* gene for methicillin resistance, *rpoB* gene for rifampin resistance). Thus, it is mainly useful in combination with conventional cultures in pretreated patients. Use of specific PCR primers is potentially more sensitive for detecting defined microorganisms. ¹¹⁰ However, in a prospective cohort study the sensitivity of a multiplex PCR in sonication fluid (Unyvero i60 [Curetis Cie, Holzgerlingen, Germany], designed for detection of PJI) was not better than that of sonicate culture (53% vs. 58%). ¹¹¹

Imaging Procedures

Radiographs have a low sensitivity and specificity for the diagnosis of PJI. Radiolucency, osteolysis, and migration are signs of not only infection but also aseptic loosening. ¹¹² Hence, the diagnostic role of radiography is limited. Ultrasonography is helpful in cases with joint effusion that cannot be clinically diagnosed and where guidance for joint aspiration is necessary. Computed tomography (CT) allows detection of soft tissue infection (abscesses, sinus tracts), prosthetic loosening, and bone erosion. With the use of special techniques, metallic artifacts can be minimized. However, this technique is more often used to estimate the extent of an infection than to determine whether or not an infection is present. Imaging interference occurs in the vicinity of metal implants. The indication for magnetic resonance imaging is similar to that for CT.

With radionuclide imaging, signs of infection are visible before anatomic changes. The three-phase bone scan using a bone-seeking tracer (e.g., technetium 99m-labeled methylene diphosphonate [^{99m}Tc-MDP]) is sensitive for detecting infection but has poor specificity. ¹¹³ Bone remodeling and hence marker uptake is increased for at least 1 year after implantation. ¹¹⁴ Furthermore, heterotopic ossification and aseptic loosening also increase tracer uptake. The specificity can be improved to 80% to 90% with use of a more specific radiotracer such as ^{99m}Tc-ciprofloxacin (Infecton) or ^{99m}Tc-antigranulocyte monoclonal antibodies. ¹¹⁵ ¹¹⁶ The use of labeled leukocytes is cumbersome and depends on the quality of the purification and labeling. ¹¹⁷ Unfortunately, the spatial resolution of nuclear scanning techniques is limited. Single-photon emission CT plus conventional CT (SPECT-CT), which is performed with an integrated hybrid machine, offers a more precise localization of the radiotracer and should therefore be preferred. This method is mainly used with ^{99m}Tc-MDP, labeled leukocytes, or labeled antigranulocyte monoclonal antibodies. It improves the sensitivity and specificity as compared with the planar image. In 31 consecutive patients with suspected low-grade PJI, sensitivity, specificity, and accuracy improved from 66%, 60%, and 61%, respectively, to 89%, 73%, and 77%, respectively. The prevalence of PJI in this population was 29%. ¹¹⁸

Positron emission tomography (PET) examination plays an important role in the detection of malignant tumors. ¹⁸ F-fluorodeoxyglucose (FDG) accumulates in cells such as neutrophils and is intracellularly phosphorylated to a stable molecule. Therefore, accumulation occurs in not only tumors but also inflammatory foci. In a prospective study including 130 patients with a painful total hip arthroplasty, the FDG-PET showed a sensitivity of 95%, a specificity of 38% and an accuracy of 56% when using MSIS criteria as gold standard. ¹¹⁹ Individual studies show sensitivities between 22.2% and 100%, and specificities between 61.5% and 100%. Reasons for these inconsistent results are the nonspecific postoperative FDG uptake after total knee arthroplasty, the prolonged increased FDG uptake around the femoral neck, and the nonspecific FDG uptake around the distal part of the prosthetic stem. ¹¹³ ¹²⁰ Thus PET cannot be recommended for routine clinical practice.

Treatment Concepts

General Aspects

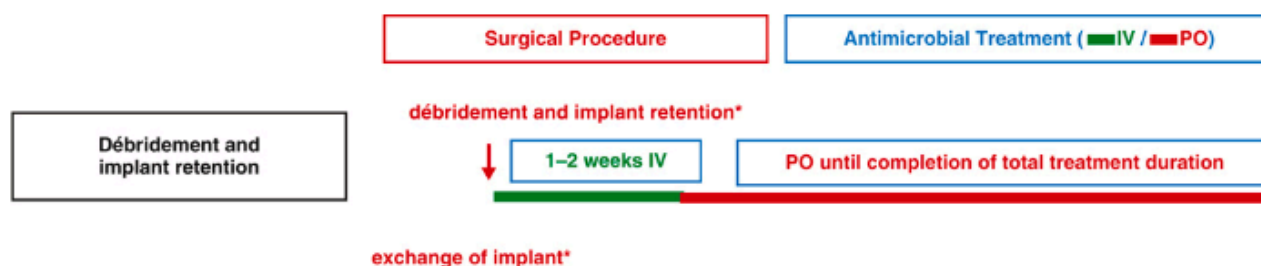
General Aspects

Cure at the first treatment attempt is crucial because with each treatment failure, tissue damage and functional integrity are worse. Delayed ambulation increases loss of muscle strength and risk of thromboembolism. Therefore, early referral to specialized centers is advised. As a rule, before starting treatment, it should be clear whether a curative or suppressive (palliative) approach will be taken. Cure is defined as a long-term pain-free functional joint with complete eradication of infection after a follow-up of 24 months after surgery. Eradication surgery requires a combination of both an appropriate surgical procedure and antimicrobial therapy. In contrast, palliative therapy is geared toward suppression of infection and thus of symptoms. In general, only minor or no surgery is performed with this strategy. Lifelong suppressive antimicrobial therapy may be necessary when the chance for cure is minimal.

The cornerstone of successful treatment is early diagnosis. The earlier the diagnosis, the less invasive is the surgical therapy. Accordingly, a high degree of suspicion is required and empirical antibiotic therapy without unequivocal diagnosis must be avoided.

Traditional standard treatment included two-stage exchange with meticulous removal of all necrotic tissue, bone cement, and the prosthesis combined with a prolonged course of antibiotics during the implant-free interval.¹²¹ During the past two decades, an algorithm for the optimal surgical therapy of the different presentations was developed at our institute.³ Choosing the least invasive treatment that cures infection is the most rational approach (see Fig. 107.1). Because data from controlled trials comparing different surgical procedures are lacking, treatment concepts vary among different centers. Recommendations are therefore based on case series and expert opinions and published in the IDSA guidelines.^{3, 12, 63} Considering the prerequisites for the successful use of each surgical procedure, all surgical interventions have a favorable outcome in more than 80% of the patients.^a

Antimicrobial treatment without any surgical intervention is not curative but only suppressive. There are three different curative options: DAIR, one-stage exchange, and two-stage exchange. In addition, in special situations, microbiologic cure without restoration of full function can be reached with removal without replacement, arthrodesis, or amputation. There are some controversies among different specialized centers: (1) One-stage exchange is less commonly performed in the United States than it is in European centers; (2) in the case of two-stage exchange, the approach with a short interval is used in only a minority of centers; and (3) some centers continue suppressive antimicrobial therapy despite curative approach, considering the small uncertainty that the infection is not cured.^{12, 126} However, the benefit of stopping treatment (e.g., reducing side effects) outweighs that uncertainty. These controversial aspects will remain until controlled trials have been conducted. Fig. 107.1 shows a treatment algorithm that allows choosing the most appropriate surgical intervention based on several variables.³ This algorithm has been developed for patients with hip and knee PJI.^{3, 53, 54, 127} In addition, it has also been implemented for shoulder and elbow prostheses.^{58, 128} In one review of shoulder periprosthetic infections, the rate of cure was 90% to 92% whether the implant was replaced in a one- or a two-stage procedure.¹²⁹ In contrast, the optimal surgical management of ankle PJI has not yet been defined.⁶² In these patients, the surgical strategy is mainly dictated by the more difficult soft tissue situation. Fig. 107.2 shows the sequence of surgical interventions and the duration of intravenous and oral antimicrobial therapy.



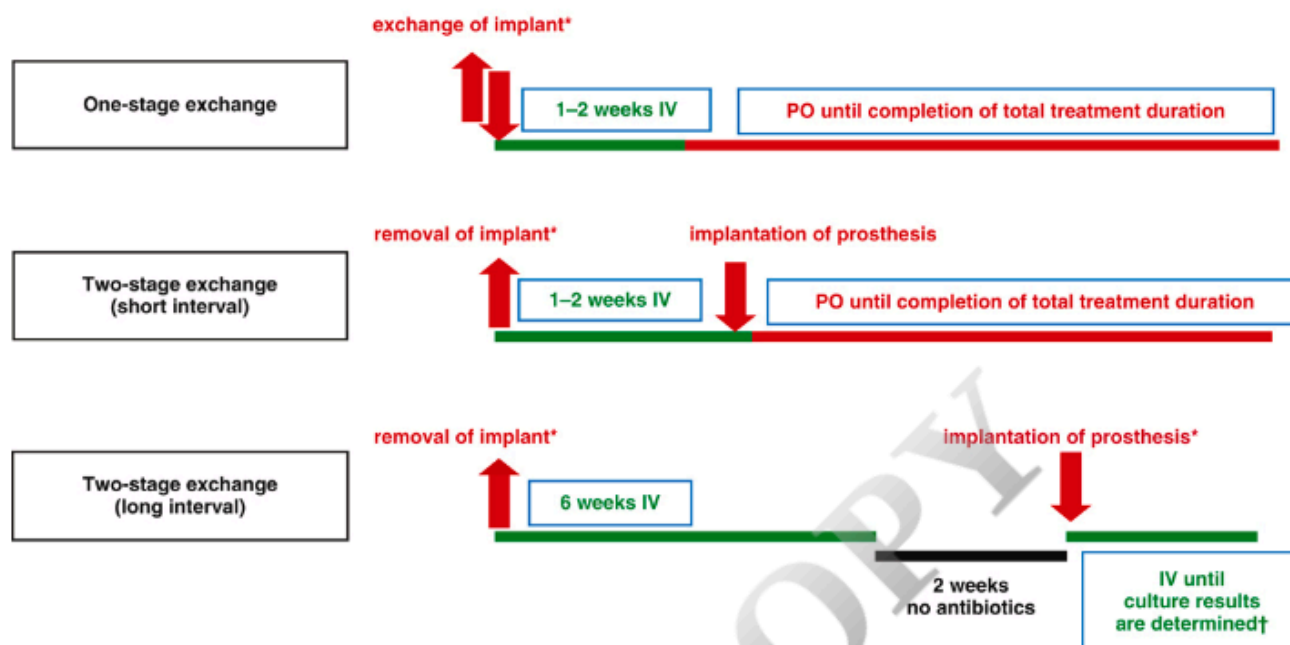


FIG. 107.2 Surgical intervention, route, and duration of antimicrobial therapy according to Zimmerli and colleagues. ³IV, Intravenous route; PO, oral route. *Microbiologic sampling. †Continuation with the same antibiotic regimen. Samples are cultured up to 14 days in centers that are specialized for periprosthetic joint infection. If microbiologic results are confirmed to be negative, the antimicrobial treatment can be stopped.

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Débridement With Implant Retention

The success rate of this procedure is traditionally widely underestimated because there are many published series that included patients with multiple risk factors for procedure failure. However, the success rate of DAIR is not lower than that of exchange surgery, provided that the following four conditions are fulfilled ^b:

- Acute infection (<1 month after implantation, or <3 weeks duration of infectious symptoms, provided that the reactivation of a latent chronic hematogenous PJI can be excluded).
- Stable implant
- Pathogen susceptible to a biofilm-active antimicrobial agent (see later)
- No sinus tract and no periprosthetic abscess

The definition for duration of symptoms is based on a controlled trial. ³⁴ Rapid and meticulous débridement of necrotic tissue is essential. In periprosthetic knee infection, the reported success rate is better after open surgery than after arthroscopic débridement (median reported success rates 86% vs. 56%). ¹³¹ In cases of open débridement, modular components should be exchanged, which results in reduction of the biofilm. The advantages of débridement with retention are the less invasive surgery and more rapid rehabilitation.

One-Stage Exchange

Direct exchange includes removal and reimplantation during the same surgical procedure. It can be chosen for patients with (1) a good soft tissue envelope and (2) a pathogen that is susceptible to oral antimicrobial agents with excellent bioavailability and activity on biofilms (see later). However, it is crucial to identify the causative pathogen before the surgical procedure. In general, effective antibiotic-impregnated bone cement is used. ¹³⁵ However, cure rate with use of

One-Stage Exchange

Direct exchange includes removal and reimplantation during the same surgical procedure. It can be chosen for patients with (1) a good soft tissue envelope and (2) a pathogen that is susceptible to oral antimicrobial agents with excellent bioavailability and activity on biofilms (see later). However, it is crucial to identify the causative pathogen before the surgical procedure. In general, effective antibiotic-impregnated bone cement is used. ¹³⁵ However, cure rate with use of uncemented revision stems is excellent in infected total hip arthroplasties. ¹³⁶

Two-Stage Exchange

The two-stage exchange procedure starts with thorough removal of all necrotic tissue, bone cement, and all of the implant before reimplantation of a new device is performed at a second intervention. ¹²⁵ After removal of all foreign material, an antibiotic-impregnated spacer, typically PMMA, is inserted in order to achieve some stability, allow some degree of mobility, prevent shrinking of the joint space, and produce a high local concentration of antibiotics. The need for local antibiotics in the spacer has never been proven in a comparative and prospective PJI trial. Interestingly, in patients with a microorganism resistant to the antibiotic in the spacer, there is no measurable Page 1372 disadvantage regarding the outcome. ¹³⁷ The advantage of staged exchange is the opportunity for removing all infected material before implantation of the new prosthesis. The disadvantage is the need for two surgical interventions, prolonged disability, and the interval with the biomechanically suboptimal spacer. Therefore, some centers favor a short interval of only 2 to 3 weeks before reimplantation, except in patients with difficult-to-treat microorganisms. ³ ¹³⁸ During the interval the bacterial load can be reduced, and the soft tissue heals. This allows treatment during one single hospitalization. Antibiotics are not stopped before implantation, and no sampling is recommended during implantation. Antibiotics are postoperatively continued for a total duration of 12 weeks.

In patients with difficult-to-treat microorganisms (small-colony variants, rifampin-resistant staphylococci, fluoroquinolone-resistant gram-negative bacilli, fungi), early reimplantation should not be chosen. ³ ¹³⁹ The rationale for the long interval between surgical procedures is the concept that difficult-to-treat microorganisms must be completely eradicated before reimplantation. Because of potential adherence and persistence of these types of microorganisms on the spacer, we recommend a two-stage procedure with a spacerless interval. ¹³⁹ We recommend that these patients be treated initially for 6 weeks after the first stage procedure, and then have a 2 week antibiotic-free period before reimplantation to enable obtainment of reliable samples for microbiologic assessment. In these patients, the same antimicrobial treatment of the previous 6 weeks should be restarted after implantation of the new device. Antibiotics can be definitively stopped when intraoperative culture results remain negative (see [Fig. 107.2](#)). If no antibiotic-free interval preceded reimplantation, a total treatment duration of 12 weeks should be chosen.

Antimicrobial Therapy

After microbiologic sampling, antimicrobial agents are administered by the intravenous route. The risk for emergence of resistance is highest during the initial phase, when the bacterial load is still high. Therefore, subinhibitory antimicrobial concentrations should be avoided. [Table 107.2](#) summarizes the pathogen-specific antimicrobial therapy, and [Fig. 107.2](#) shows the timing of the intravenous and oral therapy. It mainly depends on the type of surgery performed. The long duration of treatment is based on the reasoning that biofilm bacteria cannot be killed by host defense alone. ⁴ Therefore, PJI reoccurs if all microorganisms are not eliminated by débridement and antimicrobial therapy. In a recent open-label, randomized, controlled noninferiority trial, 6 weeks of therapy were compared to 12 weeks in patients with hip- or knee-PJI. ¹⁴⁰ The surgical procedures were débridement with implant retention (40%), one-stage exchange (39%), and two-stage exchange (21%). A total of 410 patients were randomized, 205 in each group. Persistent infection occurred in 18.1% in the 6-week group and 9.4% in the 12-week group (risk difference, 8.7%, 95% confidence interval [CI], 1.8%–15.6%). This confirms

(21%). A total of 410 patients were randomized, 205 in each group. Persistent infection occurred in 18.1% in the 6-week group and 9.4% in the 12-week group (risk difference, 8.7%, 95% confidence interval [CI], 1.8%–15.6%). This confirms that a 12-week regimen should be favored.

Initial intravenous therapy for 1 to 2 weeks is followed by long-term oral antibiotics. In a randomized controlled trial, oral (<7 days of IV-therapy) was not inferior to 6 weeks of IV therapy for different bone and joint infections. Thus, prolonged IV-therapy is not needed, as long as oral compounds with good bioavailability and reasonable bone penetration are used. ¹⁴¹ Optimal therapy is best defined in staphylococcal implant infections. Rifampin has excellent activity on susceptible biofilm staphylococci. This has been proven in vitro, in animal models, and in several clinical studies. ^c There are no clinical studies proving superiority of rifampin combinations in other gram-positive or gram-negative PJIs. ³³ Rifampin must always be combined with another drug for prevention of emergence of resistance. Owing to the risk of superinfection with rifampin-resistant staphylococci from the skin microbiome, we propose starting rifampin combination only in patients with dry wounds—that is, several days after definitive surgery. ^{36, 143} In addition, in the short-interval two-stage exchange concept, we start rifampin only after reimplantation of the device. Fluoroquinolones are the first choice for the combination. ^{3, 12, 34-36} Because of increasing resistance, other combinations may be necessary. For each combination, possible interactions with rifampin should be considered. ¹⁴⁴ Rifampin should be reserved for patients with implant retention, with one-stage exchange, or after early reimplantation in two-stage exchange. It has no advantage in patients with suppressive therapy and should therefore be discouraged with this type of treatment. Diligent use of rifampin lowers the risk for emergence of resistance. ¹⁴⁵

Because the susceptibility of methicillin-resistant staphylococci to vancomycin is decreasing, novel antistaphylococcal drugs, such as linezolid and daptomycin, may be necessary for the treatment of PJI. ^{134, 146} Notably, neither is approved for bone and joint infections. Linezolid surpasses the minimal inhibitory concentration in 90% of isolates (MIC₉₀) of enterococci and staphylococci and has a good penetration in human bone. ¹⁴⁷ Despite several observational studies in humans, linezolid efficacy in PJI cannot be unambiguously judged. ¹⁴⁸⁻¹⁵⁰ According to an observational noncomparative study, in patients with implant retention, monotherapy had a lower cure rate than combination therapy with rifampin. Notably, the best cure rate with implant retention was observed in patients with acute infections and combination therapy. ¹⁵¹ Serious adverse events of long-term linezolid must be considered, including peripheral neuropathy, optic neuritis, and bone marrow suppression. Because bone and joint infections generally require prolonged treatment, the use of linezolid remains controversial.

Daptomycin monotherapy has a low cure rate in animal models of implant-associated infections. However, in combination with rifampin it was highly efficacious in animal models of implant-associated infections. ^{152, 153} Moreover, in contrast to vancomycin, daptomycin combination therapy completely prevented the emergence of rifampin resistance in *S. aureus*. ¹⁵² However, the efficacy of daptomycin for PJI cannot be conclusively judged because there are not enough clinical data. If daptomycin is considered for selected cases, the available data point toward treatment regimens with doses ranging from 8 mg/kg up to 10 mg/kg once daily and combination with rifampin. ¹⁵⁴ During prolonged treatment, creatinine phosphokinase surveillance is necessary to rapidly detect muscle toxicity. Comedication with statins should be avoided. Also, in case of fever and dyspnea, eosinophilic pneumonia, a rare but life-threatening complication in patients with daptomycin therapy, should be considered. ¹⁵⁵

PJI caused by gram-negative bacilli is diagnosed in 10% to 15% of the cases. The risk for these microorganisms is higher in previously treated patients. ¹⁵⁶ Patients with gram-negative PJI who qualify for débridement and retention have a good outcome, provided that treatment with a fluoroquinolone is possible, after initial intravenous therapy (see **Table 107.2**). ^{37, 156-158} The rationale for favoring fluoroquinolones is their activity against gram-negative biofilms. ^{22, 23, 37, 158}

outcome, provided that treatment with a fluoroquinolone is possible, after initial intravenous therapy (see [Table 107.2](#)). [37](#), [156](#)- [158](#) The rationale for favoring fluoroquinolones is their activity against gram-negative biofilms. [22](#), [23](#), [37](#), [158](#)

Internal Fixation–Associated Infection

Metal hardware for internal fixation of bone fractures, spinal fusions, and corrections of scoliosis are included in this category. Management of PJI and internal fixation–associated infections have important differences. First, eradication of infection is not always a priority, because the device could be removed after fracture healing or because bridging until definitive stabilization is possible with an external fixation device. That stated, removal of hardware spine implants is often not feasible because of risk of spinal instability. Second, exogenous infection after surgical treatment for fractures, in particular open fractures, plays a more important role than in PJI. Third, there is a large variety of anatomic locations where orthopedic devices are fixed, and a large variety of hardware is implanted (e.g., plates, screws, nails, rods), illustrating heterogeneous clinical constellations. Fourth, the involvement of a plastic surgeon for soft tissue coverage after infection in trauma cases plays a more important role than in PJI. Consequently, the definition, classification and management concepts differ between PJI and internal fixation–associated infections.

Definition

There are no commonly accepted criteria for the definition of internal fixation–associated infections, because of the aforementioned heterogeneity of metal work used (e.g., spine implants vs. osteosynthesis after fractures). For implant-associated infection related to fractures, [Page 1373](#) however, the term *fracture-related infection* (FRI) should be used. A consensus definition for FRI was proposed in 2018, [159](#) updated in 2020, [160](#) and validated in a retrospective multicenter study. [161](#) The definition uses suggestive and confirmatory criteria. [159](#)- [161](#) A combination of clinical, laboratory, imaging, microbiologic, and histopathologic criteria is used to diagnose FRI. The sensitivity of the histopathologic criterion is 80% when using a cut-off ≥ 5 neutrophils per HPF. [162](#) Significant growth of microorganisms in sonicate or conventional culture of the device, or growth of the same microorganisms in at least two biopsy samples in proximity to the device, defines infection. In addition to other criteria, a single biopsy sample with growth of a virulent pathogen (e.g., *S. aureus*) may be sufficient to fulfill infection definition. Additional signs of infection are bone loss around metal screws or an intramedullary rod (e.g., nonunion), fluid accumulation around the device, or a draining sinus. [161](#)

TABLE 107.2

Treatment of Implant-Associated Infections

MICROORGANISM	ANTIMICROBIAL AGENT a	DOSE	ROUTE
<i>Staphylococcus</i> spp.	Recommendation for the initial IV treatment phase (for 1–2 wk)		
Methicillin susceptible	Rifampin <i>plus</i>	450 mg q12h or 600 mg q24h b	PO/IV
	Nafcillin <i>or</i> oxacillin c	2 g q4h	IV
Methicillin resistant	Rifampin <i>plus</i>	450 mg q12h or 600 mg q24h b	PO/IV
	Vancomycin <i>or</i>	15 mg/kg q12h d	IV
	Daptomycin	8–10 mg/kg q24h e	IV

Methicillin resistant	Ritampin <i>plus</i>	450 mg q12h or 600 mg q24h ^b	PO/IV
	Vancomycin or	15 mg/kg q12h ^d	IV
	Daptomycin	8–10 mg/kg q24h ^e	IV
Staphylococcus spp.	Recommendation after completion of the initial IV treatment phase		
	Rifampin plus	450 mg q12h or 600 mg q24h ^b	PO
	Levofloxacin or	750 mg q24h to 500 mg q12h	PO
	Ciprofloxacin or	750 mg q12h	PO
	Teicoplanin or	400 mg q24h ^f	IV
	Fusidic acid or	500 mg q8h	PO
	Trimethoprim-sulfamethoxazole or	1 DS tablet q8h ^g	PO
	Minocycline ^h or	100 mg q12h	PO
	Linezolid or	600 mg q12h	PO
	Clindamycin ⁱ	1200–1350 mg/day divided in 3–4 doses	PO
Streptococcus spp. ^j	Penicillin G ^c or	18–24 million U/day divided in 6 doses	IV
	Ceftriaxone for 4 wk, followed by	2 g q24h	IV
	Amoxicillin or	1000 mg q8h	PO
	Clindamycin ⁱ	1200–1350 mg/day divided in 3–4 doses	PO
Enterococcus spp. ^k			
Ampicillin or amoxicillin susceptible			IV
	Ampicillin or amoxicillin ^l or	2 g q4h–q6h	IV
Ampicillin or amoxicillin resistant	Vancomycin or	15 mg/kg q12h ^d	IV
	Daptomycin or	6–10 mg/kg q24h ^e	IV
	Linezolid	600 mg q12h	IV/PO
Enterobacteriaceae	β-Lactam based on in vitro susceptibilities for 2 wk ^m followed by		IV
	Ciprofloxacin	750 mg q12h	PO
Enterobacter spp. ⁿ	Cefepime or	2 g q8h	IV

	Trimethoprim-sulfamethoxazole <i>or</i>	1 DS tablet q8h ^g	PO
	Minocycline ^h <i>or</i>	100 mg q12h	PO
	Linezolid <i>or</i>	600 mg q12h	PO
	Clindamycin ⁱ	1200–1350 mg/day divided in 3–4 doses	PO
<i>Streptococcus</i> spp. ^j	Penicillin G ^c <i>or</i>	18–24 million U/day divided in 6 doses	IV
	Ceftriaxone for 4 wk, <i>followed by</i>	2 g q24h	IV
	Amoxicillin <i>or</i>	1000 mg q8h	PO
	Clindamycin ⁱ	1200–1350 mg/day divided in 3–4 doses	PO
<i>Enterococcus</i> spp. ^k			
Ampicillin or amoxicillin susceptible			IV
	Ampicillin or amoxicillin ^l <i>or</i>	2 g q4h–q6h	IV
Ampicillin or amoxicillin resistant	Vancomycin <i>or</i>	15 mg/kg q12h ^d	IV
	Daptomycin <i>or</i>	6–10 mg/kg q24h ^e	IV
	Linezolid	600 mg q12h	IV/PO
Enterobacteriaceae	β-Lactam based on in vitro susceptibilities for 2 wk ^m <i>followed by</i>		IV
	Ciprofloxacin	750 mg q12h	PO
<i>Enterobacter</i> spp. ⁿ	Cefepime <i>or</i>	2 g q8h	IV
	Ertapenem <i>or</i>	1 g q24h	IV
	Meropenem for 2–4 wk, <i>followed by</i>	1 g q8h	IV
	Ciprofloxacin	750 mg q12h	PO
Nonfermenters ^p (e.g., <i>Pseudomonas aeruginosa</i>)	Cefepime <i>or</i>	2 g q8h	IV
	Ceftazidime <i>or</i>	2 g q8h	IV
	Meropenem for 2–4 wk, <i>followed by</i>	1 g q8h ^o , ^p	IV
	Ciprofloxacin	750 mg q12h	PO
<i>Cutibacterium</i> spp.	Penicillin G <i>or</i>	18–24 million U/day divided in 6 doses	IV

		divided in 3–4 doses	
<i>Enterococcus</i> spp. k			
Ampicillin or amoxicillin susceptible			IV
	Ampicillin or amoxicillin l or	2 g q4h–q6h	IV
Ampicillin or amoxicillin resistant	Vancomycin or	15 mg/kg q12h d	IV
	Daptomycin or	6–10 mg/kg q24h e	IV
	Linezolid	600 mg q12h	IV/PO
Enterobacteriaceae	β -Lactam based on in vitro susceptibilities for 2 wk m followed by		IV
	Ciprofloxacin	750 mg q12h	PO
<i>Enterobacter</i> spp. n	Cefepime or	2 g q8h	IV
	Ertapenem or	1 g q24h	IV
	Meropenem for 2–4 wk, followed by	1 g q8h	IV
	Ciprofloxacin	750 mg q12h	PO
Nonfermenters p (e.g., <i>Pseudomonas aeruginosa</i>)	Cefepime or	2 g q8h	IV
	Ceftazidime or	2 g q8h	IV
	Meropenem for 2–4 wk, followed by	1 g q8h o , p	IV
	Ciprofloxacin	750 mg q12h	PO
<i>Cutibacterium</i> spp.	Penicillin G or	18–24 million U/day divided in 6 doses	IV
	Clindamycin i for 2–4 wk followed by	600–900 mg q8h	IV
	Amoxicillin or	1000 mg q8h	PO
	Clindamycin i	1200–1350 mg/day divided in 3–4 doses	PO
Gram-negative anaerobes (e.g., <i>Bacteroides</i> spp.)	Metronidazole	500 mg q8h	IV/PO
Mixed infections (without methicillin-resistant staphylococci)	Ampicillin-sulbactam or	3 g q6h	IV
	Amoxicillin-clavulanate q or	2.2 g q6h	IV
	Piperacillin-tazobactam or	4.5 g q6h–q8h	IV
	Imipenem or	500 mg q6h–q8h	IV

	2 wk ^m followed by		
	Ciprofloxacin	750 mg q12h	PO
Enterobacter spp. ⁿ	Cefepime <i>or</i>	2 g q8h	IV
	Ertapenem <i>or</i>	1 g q24h	IV
	Meropenem for 2–4 wk, followed by	1 g q8h	IV
	Ciprofloxacin	750 mg q12h	PO
Nonfermenters ^p (e.g., <i>Pseudomonas aeruginosa</i>)	Cefepime <i>or</i>	2 g q8h	IV
	Ceftazidime <i>or</i>	2 g q8h	IV
	Meropenem for 2–4 wk, followed by	1 g q8h ^o ^p	IV
	Ciprofloxacin	750 mg q12h	PO
Cutibacterium spp.	Penicillin G <i>or</i>	18–24 million U/day divided in 6 doses	IV
	Clindamycin ⁱ for 2–4 wk followed by	600–900 mg q8h	IV
	Amoxicillin <i>or</i>	1000 mg q8h	PO
	Clindamycin ⁱ	1200–1350 mg/day divided in 3–4 doses	PO
Gram-negative anaerobes (e.g., <i>Bacteroides</i> spp.)	Metronidazole	500 mg q8h	IV/PO
Mixed infections (without methicillin-resistant staphylococci)	Ampicillin-sulbactam <i>or</i>	3 g q6h	IV
	Amoxicillin-clavulanate ^q <i>or</i>	2.2 g q6h	IV
	Piperacillin-tazobactam <i>or</i>	4.5 g q6h–q8h	IV
	Imipenem <i>or</i>	500 mg q6h–q8h	IV
	Meropenem for 2–4 wk, followed by individual regimens according to antimicrobial susceptibility	1 g q8h ^o	IV

AUC_{0-24} , Area under the time-concentration curve from time 0 to 24 hours; *DS*, double-strength; *IV*, intravenous; *MIC*, minimal inhibitory concentration.

Antimicrobial dosage recommendations are based on normal renal and hepatic function. Antimicrobials should be chosen based on in vitro susceptibility, patient drug allergies and intolerances, and potential drug interactions or contraindications to a specific antimicrobial.

	Meropenem for 2–4 wk, followed by	1 g q8h ^{o, p}	IV
	Ciprofloxacin	750 mg q12h	PO
Cutibacterium spp.	Penicillin G <i>or</i>	18–24 million U/day divided in 6 doses	IV
	Clindamycin ⁱ for 2–4 wk followed by	600–900 mg q8h	IV
	Amoxicillin <i>or</i>	1000 mg q8h	PO
	Clindamycin ⁱ	1200–1350 mg/day divided in 3–4 doses	PO
Gram-negative anaerobes (e.g., Bacteroides spp.)	Metronidazole	500 mg q8h	IV/PO
Mixed infections (without methicillin-resistant staphylococci)	Ampicillin-sulbactam <i>or</i>	3 g q6h	IV
	Amoxicillin-clavulanate ^q <i>or</i>	2.2 g q6h	IV
	Piperacillin-tazobactam <i>or</i>	4.5 g q6h–q8h	IV
	Imipenem <i>or</i>	500 mg q6h–q8h	IV
	Meropenem for 2–4 wk, followed by individual regimens according to antimicrobial susceptibility	1 g q8h ^o	IV

AUC_{0-24} , Area under the time-concentration curve from time 0 to 24 hours; *DS*, double-strength; *IV*, intravenous; *MIC*, minimal inhibitory concentration.

Antimicrobial dosage recommendations are based on normal renal and hepatic function. Antimicrobials should be chosen based on in vitro susceptibility, patient drug allergies and intolerances, and potential drug interactions or contraindications to a specific antimicrobial.

^a For total duration of antimicrobial treatment, see text.

^b Other dosages and intervals of administration have been reported with equivalent success rates. ¹⁴³ Rifampin is only started when the wound is dry. ³⁵

^c In patients with delayed hypersensitivity, cefazolin (2 g q8h IV) can be administered. In patients with immediate hypersensitivity, penicillin should be replaced by vancomycin.

^d Recommended doses are based on AUC_{0-24}/MIC and trough levels. Trough levels should be monitored for nephrotoxicity. ²¹⁶

^e Recommended dose according to Infectious Diseases Society of America (IDSA) guidelines. ¹² However, doses up to 10 mg/kg q24h are reported. ¹⁵⁴

Gram-negative anaerobes (e.g., <i>Bacteroides</i> spp.)	Metronidazole	500 mg q8h	IV/PO
Mixed infections (without methicillin-resistant staphylococci)	Ampicillin-sulbactam <i>or</i>	3 g q6h	IV
	Amoxicillin-clavulanate ^a <i>or</i>	2.2 g q6h	IV
	Piperacillin-tazobactam <i>or</i>	4.5 g q6h–q8h	IV
	Imipenem <i>or</i>	500 mg q6h–q8h	IV
	Meropenem for 2–4 wk, followed by individual regimens according to antimicrobial susceptibility	1 g q8h ^o	IV

AUC_{0-24} , Area under the time-concentration curve from time 0 to 24 hours; DS, double-strength; IV, intravenous; MIC, minimal inhibitory concentration.

Antimicrobial dosage recommendations are based on normal renal and hepatic function. Antimicrobials should be chosen based on in vitro susceptibility, patient drug allergies and intolerances, and potential drug interactions or contraindications to a specific antimicrobial.

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^e Recommended dose according to Infectious Diseases Society of America (IDSA) guidelines. ¹² However, doses up to 10 mg/kg q24h are reported. ¹⁵⁴

^f Teicoplanin: Loading dose (days 1–3 of treatment) 800 mg q24h is recommended. Not available in the United States.

^g Double strength = trimethoprim 160 mg plus sulfamethoxazole 800 mg.

^h Lack of data on bone penetration. Alternatively, doxycycline 100 mg q12h PO is possible. ³³

ⁱ Higher doses PO (e.g., up to 2400 mg/day) are possible but frequently not tolerated owing to side effects.

^j We recommend determination of minimal inhibitory concentration for penicillin.

^k Combination therapy with an aminoglycoside is optional because its superiority to monotherapy in PJI is unproven. ²¹⁷ When using combination therapy, monitor for signs of ototoxicity and nephrotoxicity of aminoglycosides; the latter is potentiated with other nephrotoxic agents (e.g., vancomycin).

^l In patients with hypersensitivity to penicillin, see treatment options for penicillin-resistant enterococci.

Page 1376 Table 107.2). However, there are no recommendations, nor are there comparative studies defining the length of treatment. In the case of early or acute hematogenous infection after an uneventful postoperative period treated with implant retention and débridement, we propose 3 months of antimicrobial therapy. In other cases, treatment duration depends on the surgical intervention and may be prolonged as long as the device is retained. Bone consolidation should be regularly evaluated and the device removed after fracture consolidation. Thus, the treatment duration can be kept to the shortest possible time.

Local Antimicrobial Therapy

In open fractures, particularly in those with high bacterial contamination, the implantation of antibiotic-loaded beads or spacers to fill dead space is common practice.^{178, 179} Although an aminoglycoside (e.g., gentamicin, tobramycin) is the standard antibiotic in spacers, virtually any antimicrobial agent can be added to spacers and beads, as long as it is heat-stable.¹⁸⁰ The rationale behind this procedure is to reduce the local bacterial load in poorly vascularized areas. Also, bead chains can be placed in difficult-to-reach bone areas. In open fracture, several studies have reported a reduction of infection rates when adjuvant antibiotic-loaded beads were used.¹⁸¹ It should also be considered that there will be a selection of bacteria with emerging resistance toward the antibiotics incorporated in beads and spacers.¹⁸² Because these materials are foreign bodies, adherence of bacteria can occur.¹⁸³ Although antibiotics are released locally, uncommonly they may have systemic unwanted effects (e.g., nephrotoxicity of aminoglycosides).¹⁸⁴ Taken together, it is judicious to rapidly reduce the dead space and local bacterial load in a contaminated open fracture. Nevertheless, antibiotic-loaded beads and spacers are foreign bodies and should be removed within a short period of time. Removal is not necessary when beads constructed from bioabsorbable material are used.¹⁸⁵ Their use may result in significant leakage, making the clinical interpretation of wound oozing difficult (i.e., leakage from dissolving beads vs. infection). Of note, it should be considered that a muscle flap likely contributes its part to local antimicrobial defense via excellent vascularization and penetration into the tissue. Rapid closure of the wound is essential to hinder the penetration of skin flora into the wound.

Surgical Interventions

It is important to note that patients with implant-associated infections can benefit from rapid referral to specialized centers. Before surgery, the following factors must be assessed: viability of the soft tissue and bone, bone stability, fracture healing, implant stability, duration of the infection, and magnitude of the soft tissue defect.¹⁶⁷ The soft tissue envelope must be well vascularized and intact to provide systemic antimicrobial agents and immune cells to the infected area. As a rule, stability of bone fragments is required for union. In patients with unstable devices, there is neither bone consolidation nor healing of infection. Therefore, such implants should be removed and replaced with either a new internal fixation device (plate or intramedullary nail) or external fixation. External fixation is preferred in cases involving difficult-to-treat microorganisms (e.g., fungi or multidrug-resistant microorganisms). The majority of early infections can be treated with DAIR. This is also true for patients with acute symptoms after an uneventful postoperative course. Débridement includes diagnostic biopsy sampling and meticulous removal of all necrotic tissue including dead bone, abscess membranes, and granulation tissue. The first intervention is generally the most important. In case of extensive tissue necrosis or hematoma formation, more than one intervention may be necessary—although the more thorough the first intervention, the fewer additional débridements are required. In nonearly infections, the decision to retain or to remove the implant depends on multiple factors, also because biofilm formation occurs on a time-dependent continuum. In patients with a delayed infection presentation, the same procedure as for early infection may be applied, although with a different antimicrobial approach (see earlier). Conceivably, shorter disease duration implies a better prognosis. In patients with a long disease duration, the implant can generally not be retained. In patients with significant skin and soft tissue defects (a priori or as a consequence of meticulous débridement), the procedure should be combined with plastic and reconstructive surgery. Uncovered soft tissue defects or delayed wound healing may result in exogenous superinfection, which is often polymicrobial. The duration of both hospitalization and

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is not hindered, but likely enhanced, when the wound is covered with a vacuum-assisted closure device. [166](#)

Pin-Tract Infection

Local inflammation surrounding fixation pins of external fixators is frequent, mainly caused by heat necrosis caused by drilling at excessive speed and power. Extension of infection to the bone is less frequent. Pin infection should be rapidly treated with local cleaning and a short course of oral antibiotics. The choice of treatment agent and duration are frequently in analogy to skin- and soft-tissue infections. According to a Cochrane review, no specific pin-site care was better than any other for reducing the risk of infection. [186](#) If deep bone infection occurs, pins will loosen. A systematic review on pin-tract infections reported a cumulative incidence of 27.4%, although with a large incidence heterogeneity among published studies. [187](#) With established pin-site bone infection, a ring sequestrum will be formed. Surgical treatment of such infections requires removing the involved pins, excising the ring sequestrum, and administering a short course of antibiotics. If bone fragments are not yet stable, new pins should be inserted at a distant site.

Prevention in Orthopedic Surgery

Perioperative Antimicrobial Prophylaxis

Arthroplasty and Closed Fractures

Timely correct antimicrobial prophylaxis reduces postoperative infections. [188](#), [189](#) The Surgical Infection Prevention Guideline Writers Workgroup has suggested the infusion of the first antimicrobial dose within 60 minutes before surgical incision. [190](#) Although several studies observed even lower infection rates when antibiotic prophylaxis was administered within 30 minutes before incision, [191](#), [192](#) others have not found statistical significant differences within the 60-minute time frame. [193](#) A controlled trial randomized 5580 patients (39% of them in a trauma unit) for two different prophylaxis time points. [194](#) The median administration times before incision were 42 (interquartile range [IQR], 30–55) versus 16 (IQR, 10–25) minutes, and the overall rate of surgical site infections was 5% in both groups. In view of surgical site infections in patients with orthopedic implants, a follow-up of only 30 days is a limitation of this trial.

Most studies investigating the prophylactic effect of antibiotic-impregnated bone cement in elective orthopedic surgery have failed to demonstrate significant differences because of the small numbers of patients. However, a study including 43,149 primary and revision knee arthroplasties (Finnish Arthroplasty Register) was able to show that the lack of antibiotic-impregnated cement increases the risk for reoperation because of infection. Cases in which patients received only intravenous prophylaxis were compared with those in which patients received combined antibiotic prophylaxis (hazard ratio [HR], 1.42 [95% CI, 1.08–1.88] for primary arthroplasty; HR, 2.12 [95% CI, 1.14–3.92] for revision arthroplasty). [195](#)

Intravenous antimicrobial prophylaxis should be stopped after the end of surgery. [196](#) Continuation does not further reduce infection rates but may increase the incidence of resistant pathogens in subsequent nosocomial infections. [197](#)–[199](#)

Compounds for antimicrobial prophylaxis are usually selected according to local epidemiology, national guidelines, and in-house practices. Most centers use first- or second-generation cephalosporins. Dual antibiotic prophylaxis (i.e., cefazolin plus vancomycin) failed to prove superiority in reducing the overall rate of surgical site infections in comparison with cefazolin monotherapy. [200](#) Nevertheless, if the local epidemiology points toward high MRSA or multiresistant gram-negative colonization rates, or both, the compounds used for antimicrobial prophylaxis may be adapted accordingly, based on the recommendations of local hospital infection control specialists. Of note, the addition of vancomycin to cefazolin in the prophylaxis regimen increases the risk of acute kidney injury. [201](#) Page 1377

In cases of suspected or definite orthopedic implant-associated infections, there are additional aspects regarding the timing of antimicrobial prophylaxis. Obtaining biopsy specimens after antimicrobial agents have reached significant tissue levels decreases the sensitivity of microbiologic culture by less than 10%. [202](#) The difference in sensitivity may not be significant in chronic infections. [202](#), [203](#) Preferentially, samples should be obtained just before administration of

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In cases of suspected or definite orthopedic implant-associated infections, there are additional aspects regarding the timing of antimicrobial prophylaxis. Obtaining biopsy specimens after antimicrobial agents have reached significant tissue levels decreases the sensitivity of microbiologic culture by less than 10%. ²⁰² The difference in sensitivity may not be significant in chronic infections. ²⁰²⁻²⁰³ Preferentially, samples should be obtained just before administration of antimicrobial prophylaxis.

Open Fractures

Open fractures are at increased risk for infection (see [Table 107.3](#)). Therefore, emergency surgery with antibiotic preemptive therapy (e.g., first-generation cephalosporins for up to 24 hours) is necessary. For type I and II fractures, the same compounds used for closed fractures are indicated. In cases of massive exogenous contamination in type III open fractures, certain experts recommend antibiotics that are more active against anaerobes or gram-negatives (e.g., ampicillin-sulbactam or piperacillin-tazobactam or flucloxacillin plus an aminoglycoside). ²⁰⁴ Most recommendations are based on institutional preferences, ²⁰⁵ as studies comparing different compounds are limited by study design and sample size. ²⁰⁶⁻²⁰⁷

In type IIIB and IIIC fractures, preemptive systemic antimicrobial therapy is recommended. Similar to the lack of data regarding the optimal compound, the optimal duration is unknown. Dellinger and colleagues ²⁰⁸ compared in a double-blind prospective trial 1 day versus 5 days of preemptive cephalosporin treatment in patients with open extremity fractures. The rates of postoperative fracture site infections were similar. Similarly, in a retrospective case-control study, Dunkel and colleagues ²⁰⁹ found no difference in subsequent infection rates when a 1-day course of antibiotics was compared with prolonged treatment, even for grade III open fractures. A systematic review concluded that available studies on preemptive therapy have methodological concerns, and that prolonged duration is not beneficial in the prevention of FRI in open fractures. ²¹⁰ We recommend a 1-day (or a maximum 3-day) preemptive therapy course with ampicillin-sulbactam or amoxicillin-clavulanic acid in patients with grade III open fractures. The use of local prophylactic antibiotics in open limb fractures is discussed above. ¹⁸¹ Closure of the wound should be strived for as soon as soft tissue conditions allow doing so in cases of open fractures. Also, rapid stabilization of the fracture is necessary. In type III fractures, initial external fixation should be preferred for a short period because of the high risk of infection with intramedullary nailing. Hence, adequate surgical procedures reduce the rate of posttraumatic infections, illustrating the importance of an interdisciplinary approach.

Prevention of Hematogenous Infection

Because implants are highly susceptible to bacterial adherence, the risk of hematogenous infection remains as long as the

Because implants are highly susceptible to bacterial adherence, the risk of hematogenous infection remains as long as the device is implanted. ^{25-27, 39, 40} Accordingly, means to prevent hematogenous infection are limited. Patients should be informed about the risk and reminded to rapidly contact a physician when signs of infection occur, irrespective of localization. It is important to treat obvious infections (e.g., pneumonia, skin and soft tissue infections, urinary tract infections) rapidly to prevent bacteremia.

A further approach is the application of hygiene measures to reduce potential sources of bacteremia. These include *S. aureus* decolonization and dental hygiene procedures. In a double-blind, placebo-controlled, randomized study including patients undergoing elective orthopedic surgery, eradication of *S. aureus* nasal carriage with mupirocin was not effective in reducing surgical site infections. However, the rate of hematogenous *S. aureus* infections was five times lower in the mupirocin group. ²¹¹ Similarly, orodental hygiene and regular dental treatment can be recommended because these might result in reduction of risk of hematogenous PJI. ²¹² In contrast, antibiotic prophylaxis to reduce the risk of hematogenous seeding and joint contamination during dental surgery is unnecessary. ²¹²⁻²¹⁴ This is true for both low- and high-risk dental procedures. ^{212, 213} Similarly, genitourinary procedures are not risk factors for subsequent PJI, and consequently antibiotic prophylaxis is not required. ²¹⁵

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References

1. Darouiche R.O. Treatment of infections associated with surgical implants. *N Engl J Med*. 2004;350(14):1422–1429.
- Trampuz A, Zimmerli W. Diagnosis and treatment of infections associated with fracture-fixation devices. *Injury*. 2006;37(Suppl 2):S59–S66.
3. Zimmerli W, Trampuz A, Ochsner P.E. Prosthetic-joint infections. *N Engl J Med*. 2004;351(16):1645–1654.
4. Zimmerli W, Sendi P. Pathogenesis of implant-associated infection: the role of the host. *Semin Immunopathol*. 2011;33(3):295–306.
5. Bernaus M, Auñón-Rubio Á, Monfort-Mira M, et al. Risk factors of DAIR failure and validation of the KLIC score: a multicenter study of four hundred fifty-five patients. *Surg Infect (Larchmt)*. 2022;23(3):280–287.
6. Vasoo S, Chan M, Sendi P, Berbari E. The value of ortho-ID teams in treating bone and joint infections. *J Bone Jt Infect*. 2019;4(6):295–299.
7. Horan T.C, Andrus M, Dudeck M.A. CDC/NHSN surveillance definition of health care-associated infection and criteria for specific types of infections in the acute care setting. *Am J Infect Control*. 2008;36(5):309–332.
8. Berbari E.F, Hanssen A.D, Duffy M.C, et al. Risk factors for prosthetic joint infection: case-control study. *Clin Infect Dis*. 1998;27(5):1247–1254.
9. Shohat N, Bauer T, Buttaro M, et al. Hip and knee section, what is the definition of a periprosthetic joint infection (PJI) of the knee and the hip? Can the same criteria be used for both joints?: proceedings of international consensus on orthopedic infections. *J Arthroplasty*. 2019;34(2s):S325–s327.
10. McNally M, Sousa R, Wouthuyzen-Bakker M, et al. The EBJIS definition of periprosthetic joint infection. *Bone Joint J*. 2021;103-b(1):18–25.
11. Parvizi J, Tan T.L, Goswami K, et al. The 2018 definition of periprosthetic hip and knee infection: an evidence-based and validated criteria. *J Arthroplasty*. 2018;33(5):1309–1314.e2.
12. Osmon D.R, Berbari E.F, Berendt A.R, et al. Diagnosis and management of prosthetic joint infection: clinical practice guidelines by the Infectious Diseases Society of America. *Clin Infect Dis*. 2013;56(1):e1–e25.
13. Pérez-Prieto D, Portillo M.E, Puig-Verdié L, et al. C-reactive protein may misdiagnose prosthetic joint infections, particularly chronic and low-grade infections. *Int Orthop*. 2017;41(7):1315–1319.

- infections, particularly chronic and low-grade infections. *Int Orthop*. 2017;41(7):1315–1319.
14. Anderson J.M, McNally A.K. Biocompatibility of implants: lymphocyte/macrophage interactions. *Semin Immunopathol*. 2011;33(3):221–233.
 15. François P, Vaudaux P, Foster T.J, Lew D.P. Host-bacteria interactions in foreign body infections. *Infect Control Hosp Epidemiol*. 1996;17(8):514–520.
 16. Bernard L, Vaudaux P, Merle C, et al. The inhibition of neutrophil antibacterial activity by ultra-high molecular weight polyethylene particles. *Biomaterials*. 2005;26(27):5552–5557.
 17. Zimmerli W, Lew P.D, Waldvogel F.A. Pathogenesis of foreign body infection. Evidence for a local granulocyte defect. *J Clin Invest*. 1984;73(4):1191–1200.
 18. Bauer T.W, Schils J. The pathology of total joint arthroplasty.II. Mechanisms of implant failure. *Skeletal Radiol*. 1999;28(9):483–497.
 19. Elek S.D, Conen P.E. The virulence of *Staphylococcus pyogenes* for man; a study of the problems of wound infection. *Br J Exp Pathol*. 1957;38(6):573–586.
 20. Widmer A.F, Frei R, Rajacic Z, Zimmerli W. Correlation between in vivo and in vitro efficacy of antimicrobial agents against foreign body infections. *J Infect Dis*. 1990;162(1):96–102.
 - Furustrand T.U, Corvec S, Betrisey B, Zimmerli W, Trampuz A. Role of rifampin against *Propionibacterium acnes* biofilm in vitro and in an experimental foreign-body infection model. *Antimicrob Agents Chemother*. 2012;56(4):1885–1891.
 22. Widmer A.F, Wiestner A, Frei R, Zimmerli W. Killing of nongrowing and adherent *Escherichia coli* determines drug efficacy in device-related infections. *Antimicrob Agents Chemother*. 1991;35(4):741–746.
 23. Widmer A.F, Colombo V.E, Gächter A, Thiel G, Zimmerli W. *Salmonella* infection in total hip replacement: tests to predict the outcome of antimicrobial therapy. *Scand J Infect Dis*. 1990;22(5):611–618.
 24. Zimmerli W, Zak O, Vosbeck K. Experimental hematogenous infection of subcutaneously implanted foreign bodies. *Scand J Infect Dis*. 1985;17(3):303–310.
 25. Murdoch D.R, Roberts S.A, Fowler Jr. V.G, et al. Infection of orthopedic prostheses after *Staphylococcus aureus* bacteremia. *Clin Infect Dis*. 2001;32(4):647–649.
 - Sendi P, Banderet F, Graber P, Zimmerli W. Periprosthetic joint infection following *Staphylococcus aureus* bacteremia. *J Infect*. 2011;63(1):17–22.
 27. Tande A.J, Palraj B.R, Osmon D.R, et al. Clinical presentation, risk factors, and outcomes of hematogenous prosthetic joint infection in patients with *Staphylococcus aureus* bacteremia. *Am J Med*. 2016;129(2):221.e11–221.e20.
 28. Kristian S.A, Birkenstock T.A, Sauder U, Mack D, Götz F, Landmann R. Biofilm formation induces C3a release and protects *Staphylococcus epidermidis* from IgG and complement deposition and from neutrophil-dependent killing. *J Infect Dis*. 2008;197(7):1028–1035.
 29. del Pozo J.L, Patel R. The challenge of treating biofilm-associated bacterial infections. *Clin Pharmacol Ther*. 2007;82(2):204–209.
 30. Darouiche R.O, Dhir A, Miller A.J, Landon G.C, Raad I.I, Musher D.M. Vancomycin penetration into biofilm covering infected prostheses and effect on bacteria. *J Infect Dis*. 1994;170(3):720–723.
 31. Lewis K. Riddle of biofilm resistance. *Antimicrob Agents Chemother*. 2001;45(4):999–1007.
 32. Zimmerli W, Frei R, Widmer A.F, Rajacic Z. Microbiological tests to predict treatment outcome in experimental device-related infections due to *Staphylococcus aureus*. *J Antimicrob Chemother*. 1994;33(5):959–967.
 33. Sendi P, Zimmerli W. Antimicrobial treatment concepts for orthopaedic device-related infection. *Clin Microbiol Infect*. 2012;18(12):1176–1184.

Microbiol Infect. 2012;18(12):1176–1184.

34. Zimmerli W, Widmer A.F, Blatter M, Frei R, Ochsner P.E. Role of rifampin for treatment of orthopedic implant-related staphylococcal infections: a randomized controlled trial. Foreign-Body Infection (FBI) Study Group. *JAMA*. 1998;279(19):1537–1541.
35. Zimmerli W, Sendi P. Role of rifampin against staphylococcal biofilm infections in vitro, in animal models, and in orthopedic-device-related infections. *Antimicrob Agents Chemother.* 2019;63(2):e01746-18.
36. Beldman M, Löwik C, Soriano A, et al. If, when, and how to use rifampin in acute staphylococcal periprosthetic joint infections, a multicentre observational study. *Clin Infect Dis.* 2021;73(9):1634–1641.
37. Rodríguez-Pardo D, Pigrau C, Lora-Tamayo J, et al. Gram-negative prosthetic joint infection: outcome of a debridement, antibiotics and implant retention approach. A large multicentre study. *Clin Microbiol Infect.* 2014;20(11):O911–O919.
38. Sendi P, Banderet F, Graber P, Zimmerli W. Clinical comparison between exogenous and haematogenous periprosthetic joint infections caused by *Staphylococcus aureus*. *Clin Microbiol Infect.* 2011;17(7):1098–1100.
39. Maderazo E.G, Judson S, Pasternak H. Late infections of total joint prostheses. A review and recommendations for prevention. *Clin Orthop Relat Res.* 1988;229:131–142.
40. Huotari K, Peltola M, Jämsen E. The incidence of late prosthetic joint infections: a registry-based study of 112,708 primary hip and knee replacements. *Acta Orthop.* 2015;86(3):321–325.
41. Löwik C.A.M, Jutte P.C, Tornero E, et al. Predicting failure in early acute prosthetic joint infection treated with debridement, antibiotics, and implant retention: external validation of the KLIC score. *J Arthroplasty.* 2018;33(8):2582–2587.
42. Mraovic B, Suh D, Jacovides C, Parvizi J. Perioperative hyperglycemia and postoperative infection after lower limb arthroplasty. *J Diabetes Sci Technol.* 2011;5(2):412–418.
43. Wouthuyzen-Bakker M, Shohat N, Parvizi J, Soriano A. Risk scores and machine learning to identify patients with acute periprosthetic joints infections that will likely fail classical irrigation and debridement. *Front Med (Lausanne).* 2021;8:550095.
44. George D.A, Drago L, Scarponi S, Gallazzi E, Haddad F.S, Romano C.L. Predicting lower limb periprosthetic joint infections: a review of risk factors and their classification. *World J Orthop.* 2017;8(5):400–411.
45. Goltz D.E, Baumgartner B.T, Politzer C.S, DiLallo M, Bolognesi M.P, Seyler T.M. The American College of surgeons national surgical quality improvement Program surgical risk calculator has a role in predicting discharge to post-acute care in total joint arthroplasty. *J Arthroplasty.* 2018;33(1):25–29.
46. Sendi P, Borens O, Wahl P, Clauss M, Uçkay I. Management of asymptomatic bacteriuria, urinary catheters and symptomatic urinary tract infections in patients undergoing surgery for joint replacement: a position paper of the expert group 'infection' of swissorthopaedics. *J Bone Jt Infect.* 2017;2(3):154–159.
47. Honkanen M, Jämsen E, Karppelin M, et al. The impact of preoperative bacteriuria on the risk of periprosthetic joint infection after primary knee or hip replacement: a retrospective study with a 1-year follow up. *Clin Microbiol Infect.* 2018;24(4):376–380.
48. Godshaw B.M, Warren M.S, Nammour M.A, Chimento G.F, Mohammed A.E, Waddell B.S. Insulin-dependent diabetic patients are at increased risk of postoperative hyperglycemia when undergoing total joint arthroplasty. *J Arthroplasty.* 2020;35(9):2375–2379.
49. Rakow A, Perka C, Trampuz A, Renz N. Origin and characteristics of haematogenous periprosthetic joint infection. *Clin Microbiol Infect.* 2019;25(7):845–850.
50. Eid A.J, Berbari E.F, Sia I.G, Wengenack N.L, Osmon D.R, Razonable R.R. Prosthetic joint infection due to rapidly growing mycobacteria: report of 8 cases and review of the literature. *Clin Infect Dis.* 2007;45(6):687–

- 694.
- Kim S.J, Kim J.H. Late onset *Mycobacterium tuberculosis* infection after total knee arthroplasty: a systematic review and pooled analysis. *Scand J Infect Dis*. 2013;45(12):907–914.
52. Marculescu C, Zimmerli W. Unusual microorganisms in periprosthetic joint infection. In: Zimmerli W, ed. *Bone and Joint Infections: From Microbiology to Diagnostics and Treatment*. 2nd ed. Hoboken, NJ: John Wiley & Sons, Inc.; 2021:21–49.
53. Giulieri S.G, Graber P, Ochsner P.E, Zimmerli W. Management of infection associated with total hip arthroplasty according to a treatment algorithm. *Infection*. 2004;32(4):222–228.
54. Laffer R.R, Graber P, Ochsner P.E, Zimmerli W. Outcome of prosthetic knee-associated infection: evaluation of 40 consecutive episodes at a single centre. *Clin Microbiol Infect*. 2006;12(5):433–439.
- Stefánsdóttir A, Johansson D, Knutson K, Lidgren L, Robertsson O. Microbiology of the infected knee arthroplasty: report from the Swedish Knee Arthroplasty Register on 426 surgically revised cases. *Scand J Infect Dis*. 2009;41(11–12):831–840.
56. Zappe B, Graf S, Ochsner P.E, Zimmerli W, Sendi P. *Propionibacterium* spp. in prosthetic joint infections: a diagnostic challenge. *Arch Orthop Trauma Surg*. 2008;128(10):1039–1046.
57. Tai D.B.G, Patel R, Abdel M.P, Berbari E.F, Tande A.J. Microbiology of hip and knee periprosthetic joint infections: a database study. *Clin Microbiol Infect*. 2022;28(2):255–259.
58. Achermann Y, Sahin F, Schwyzer H.K, Kolling C, Wüst J, Vogt M. Characteristics and outcome of 16 periprosthetic shoulder joint infections. *Infection*. 2013;41(3):613–620.
59. Tande A.J, Patel R. Prosthetic joint infection. *Clin Microbiol Rev*. 2014;27(2):302–345.
60. Marculescu C.E, Cantey J.R. Polymicrobial prosthetic joint infections: risk factors and outcome. *Clin Orthop Relat Res*. 2008;466(6):1397–1404.
61. Browning S, Manning L, Metcalf S, et al. Characteristics and outcomes of culture-negative prosthetic joint infections from the Prosthetic Joint Infection in Australia and New Zealand Observational (PIANO) cohort study. *J Bone Joint Infect*. 2022;7(5):203–211.
62. Kessler B, Knupp M, Graber P, et al. The treatment and outcome of peri-prosthetic infection of the ankle: a single cohort-centre experience of 34 cases. *Bone Joint J*. 2014;96-b(6):772–777.
63. Del Pozo J.L, Patel R. Clinical practice. Infection associated with prosthetic joints. *N Engl J Med*. 2009;361(8):787–794.
64. Sendi P, Löttscher P.O, Kessler B, Graber P, Zimmerli W, Clauss M. Debridement and implant retention in the management of hip periprosthetic joint infection: outcomes following guided and rapid treatment at a single centre. *Bone Joint J*. 2017;99-b(3):330–336.
65. Beutler A.M, Epstein A.L, Policastro D. Acute gouty arthritis involving a prosthetic knee joint. *J Clin Rheumatol*. 2000;6(5):291–293.
66. Renz N, Trampuz A, Perka C, Rakow A. Outcome and failure analysis of 132 episodes of hematogenous periprosthetic joint infections-A cohort study. *Open Forum Infect Dis*. 2022;9(4):ofac094.
67. Chang M.J, Ro D.H, Kim T.W, et al. Worse outcome of debridement, antibiotics, and implant retention in acute hematogenous infections than in postsurgical infections after total knee arthroplasty: a multicenter study. *Knee Surg Relat Res*. 2022;34(1):38.
68. Gawkrödger D.J. Metal sensitivities and orthopaedic implants revisited: the potential for metal allergy with the new metal-on-metal joint prostheses. *Br J Dermatol*. 2003;148(6):1089–1093.
69. Cousen P.J, Gawkrödger D.J. Metal allergy and second-generation metal-on-metal arthroplasties. *Contact Dermatitis*. 2012;66(2):55–62.
70. Portillo M.F, Salvadó M, Alier A, et al. Prosthesis failure within 2 years of implantation is highly predictive of

71. Berbari E, Mabry T, Tsaras G, et al. Inflammatory blood laboratory levels as markers of prosthetic joint infection: a systematic review and meta-analysis. *J Bone Joint Surg Am.* 2010;92(11):2102–2109.
72. Rienmüller A, Borens O. *Propionibacterium* prosthetic joint infection: experience from a retrospective database analysis. *Eur J Orthop Surg Traumatol.* 2016;26(4):429–434.
73. Deirmengian C.A, Citrano P.A, Gulati S, Kazarian E.R, Stave J.W, Kardos K.W. The C-reactive protein may not detect infections caused by less-virulent organisms. *J Arthroplasty.* 2016;31(9 Suppl):152–155. Page 1378.e2.
74. Huang J, Zhang Y, Wang Z, et al. The serum level of D-Dimer is not suitable for distinguishing between prosthetic joint infection and aseptic loosening. *J Orthop Surg Res.* 2019;14(1):407.
75. Worthington T, Dunlop D, Casey A, Lambert R, Luscombe J, Elliott T. Serum procalcitonin, interleukin-6, soluble intercellular adhesion molecule-1 and IgG to short-chain exocellular lipoteichoic acid as predictors of infection in total joint prosthesis revision. *Br J Biomed Sci.* 2010;67(2):71–76.
76. Drago L, Vassena C, Dozio E, et al. Procalcitonin, C-reactive protein, interleukin-6, and soluble intercellular adhesion molecule-1 as markers of postoperative orthopaedic joint prosthesis infections. *Int J Immunopathol Pharmacol.* 2011;24(2):433–440.
77. Hunziker S, Hügler T, Schuchardt K, et al. The value of serum procalcitonin level for differentiation of infectious from noninfectious causes of fever after orthopaedic surgery. *J Bone Joint Surg Am.* 2010;92(1):138–148.
78. Schinsky M.F, Della Valle C.J, Sporer S.M, Paprosky W.G. Perioperative testing for joint infection in patients undergoing revision total hip arthroplasty. *J Bone Joint Surg Am.* 2008;90(9):1869–1875.
79. Trampuz A, Hanssen A.D, Osmon D.R, Mandrekar J, Steckelberg J.M, Patel R. Synovial fluid leukocyte count and differential for the diagnosis of prosthetic knee infection. *Am J Med.* 2004;117(8):556–562.
80. Ghanem E, Parvizi J, Burnett R.S, et al. Cell count and differential of aspirated fluid in the diagnosis of infection at the site of total knee arthroplasty. *J Bone Joint Surg Am.* 2008;90(8):1637–1643.
81. Cipriano C.A, Brown N.M, Michael A.M, Moric M, Sporer S.M, Della Valle C.J. Serum and synovial fluid analysis for diagnosing chronic periprosthetic infection in patients with inflammatory arthritis. *J Bone Joint Surg Am.* 2012;94(7):594–600.
82. Bedair H, Ting N, Jacovides C, et al. The Mark Coventry Award: diagnosis of early postoperative TKA infection using synovial fluid analysis. *Clin Orthop Relat Res.* 2011;469(1):34–40.
83. Deirmengian C, Kardos K, Kilmartin P, Gulati S, Citrano P, Booth Jr. R.E. The alpha-defensin test for periprosthetic joint infection responds to a wide spectrum of organisms. *Clin Orthop Relat Res.* 2015;473(7):2229–2235.
84. Frangiamore S.J, Saleh A, Grosso M.J, et al. α -Defensin as a predictor of periprosthetic shoulder infection. *J Shoulder Elbow Surg.* 2015;24(7):1021–1027.
85. Bonanzinga T, Zahar A, Dütsch M, Lausmann C, Kendoff D, Gehrke T. How reliable is the alpha-defensin immunoassay test for diagnosing periprosthetic joint infection? A prospective study. *Clin Orthop Relat Res.* 2017;475(2):408–415.
86. Shahi A, Parvizi J, Kazarian G.S, et al. The alpha-defensin test for periprosthetic joint infections is not affected by prior antibiotic administration. *Clin Orthop Relat Res.* 2016;474(7):1610–1615.
87. Kasperek M.F, Kasperek M, Boettner F, Faschingbauer M, Hahne J, Dominkus M. Intraoperative diagnosis of periprosthetic joint infection using a novel alpha-defensin lateral flow assay. *J Arthroplasty.* 2016;31(12):2871–2874.
88. Sigmund I.K, Holinka J, Gamper J, et al. Qualitative α -defensin test (Synovasure) for the diagnosis of periprosthetic infection in revision total joint arthroplasty. *Bone Joint J.* 2017;99-b(1):66–72.

88. Sigmund I.K, Holinka J, Gamper J, et al. Qualitative α -defensin test (Synovasure) for the diagnosis of periprosthetic infection in revision total joint arthroplasty. *Bone Joint J.* 2017;99-b(1):66–72.
89. Renz N, Yermak K, Perka C, Trampuz A. Alpha defensin lateral flow test for diagnosis of periprosthetic joint infection: not a screening but a confirmatory test. *J Bone Joint Surg Am.* 2018;100(9):742–750.
90. Deirmengian C, Kardos K, Kilmartin P, et al. The alpha-defensin test for periprosthetic joint infection outperforms the leukocyte esterase test strip. *Clin Orthop Relat Res.* 2015;473(1):198–203.
91. Karbysheva S, Yermak K, Grigoricheva L, Renz N, Perka C, Trampuz A. Synovial fluid d-lactate-A novel pathogen-specific biomarker for the diagnosis of periprosthetic joint infection. *J Arthroplasty.* 2020;35(8):2223–2229.e2.
92. Widmer A.F. New developments in diagnosis and treatment of infection in orthopedic implants. *Clin Infect Dis.* 2001;33(Suppl 2):S94–S106.
93. Gallo J, Kolar M, Dendis M, et al. Culture and PCR analysis of joint fluid in the diagnosis of prosthetic joint infection. *New Microbiol.* 2008;31(1):97–104.
94. Font-Vizcarra L, García S, Martínez-Pastor J.C, Sierra J.M, Soriano A. Blood culture flasks for culturing synovial fluid in prosthetic joint infections. *Clin Orthop Relat Res.* 2010;468(8):2238–2243.
95. Atkins B.L, Athanasou N, Deeks J.J, et al. Prospective evaluation of criteria for microbiological diagnosis of prosthetic-joint infection at revision arthroplasty. The OSIRIS Collaborative Study Group. *J Clin Microbiol.* 1998;36(10):2932–2939.
96. Bémer P, Léger J, Tandé D, et al. How many samples and how many culture media to diagnose a prosthetic joint infection: a clinical and microbiological prospective multicenter study. *J Clin Microbiol.* 2016;54(2):385–391.
- Blackmur J.P, Tang E.Y, Dave J, Simpson A.H. Use of broth cultures peri-operatively to optimise the microbiological diagnosis of musculoskeletal implant infections. *Bone Joint J.* 2014;96-b(11):1566–1570.
98. Peel T.N, Spelman T, Dylla B.L, et al. Optimal periprosthetic tissue specimen number for diagnosis of prosthetic joint infection. *J Clin Microbiol.* 2017;55(1):234–243.
99. Bossard D.A, Ledergerber B, Zingg P.O, et al. Optimal length of cultivation time for isolation of *Propionibacterium acnes* in suspected bone and joint infections is more than 7 days. *J Clin Microbiol.* 2016;54(12):3043–3049.
100. Minassian A.M, Newnham R, Kalimeris E, Bejon P, Atkins B.L, Bowler I.C. Use of an automated blood culture system (BD BACTEC™) for diagnosis of prosthetic joint infections: easy and fast. *BMC Infect Dis.* 2014;14:233.
101. Bori G, Muñoz-Mahamud E, Garcia S, et al. Interface membrane is the best sample for histological study to diagnose prosthetic joint infection. *Mod Pathol.* 2011;24(4):579–584.
102. Tohtz S.W, Müller M, Morawietz L, Winkler T, Perka C. Validity of frozen sections for analysis of periprosthetic loosening membranes. *Clin Orthop Relat Res.* 2010;468(3):762–768.
103. Tunney M.M, Patrick S, Curran M.D, et al. Detection of prosthetic hip infection at revision arthroplasty by immunofluorescence microscopy and PCR amplification of the bacterial 16S rRNA gene. *J Clin Microbiol.* 1999;37(10):3281–3290.
104. Trampuz A, Piper K.E, Jacobson M.J, et al. Sonication of removed hip and knee prostheses for diagnosis of infection. *N Engl J Med.* 2007;357(7):654–663.
105. Portillo M.E, Salvadó M, Sorli L, et al. Multiplex PCR of sonication fluid accurately differentiates between prosthetic joint infection and aseptic failure. *J Infect.* 2012;65(6):541–548.
106. Liu H, Zhang Y, Li L, Zou H.C. The application of sonication in diagnosis of periprosthetic joint infection. *Eur J Clin Microbiol Infect Dis.* 2017;36(1):1–9.

- infection. *Eur J Clin Microbiol Infect Dis*. 2017;36(1):1–9.
107. Rak M, Kavčič M, Trebše R, ČóR A. Detection of bacteria with molecular methods in prosthetic joint infection: sonication fluid better than periprosthetic tissue. *Acta Orthop*. 2016;87(4):339–345.
108. Portillo M.E, Corvec S. Identification of pathogens in bone and joint infections by non-culture techniques. In: Zimmerli W, ed. *Bone and Joint Infections: From Microbiology to Diagnostics and Treatment*. 2nd ed. Hoboken, NJ: John Wiley & Sons, Inc.; 2021:51–64.
109. Achermann Y, Vogt M, Leunig M, Wüst J, Trampuz A. Improved diagnosis of periprosthetic joint infection by multiplex PCR of sonication fluid from removed implants. *J Clin Microbiol*. 2010;48(4):1208–1214.
110. Levy P.Y, Fournier P.E, Fenollar F, Raoult D. Systematic PCR detection in culture-negative osteoarticular infections. *Am J Med*. 2013;126(12):1143.e25–1143.e33.
111. Renz N, Feihl S, Cabric S, Trampuz A. Performance of automated multiplex PCR using sonication fluid for diagnosis of periprosthetic joint infection: a prospective cohort. *Infection*. 2017;45(6):877–884.
112. Stumpe K.D, Nötzli H.P, Zanetti M, et al. FDG PET for differentiation of infection and aseptic loosening in total hip replacements: comparison with conventional radiography and three-phase bone scintigraphy. *Radiology*. 2004;231(2):333–341.
113. Gemmel F, Van den Wyngaert H, Love C, Welling M.M, Gemmel P, Palestro C.J. Prosthetic joint infections: radionuclide state-of-the-art imaging. *Eur J Nucl Med Mol Imaging*. 2012;39(5):892–909.
114. Utz J.A, Lull R.J, Galvin E.G. Asymptomatic total hip prosthesis: natural history determined using Tc-99m MDP bone scans. *Radiology*. 1986;161(2):509–512.
115. Fuster D, Soriano A, Garcia S, et al. Usefulness of 99mTc-ciprofloxacin scintigraphy in the diagnosis of prosthetic joint infections. *Nucl Med Commun*. 2011;32(1):44–51.
116. Ivančević V, Perka C, Hasart O, Sandrock D, Munz D.L. Imaging of low-grade bone infection with a technetium-99m labelled monoclonal anti-NCA-90 Fab' fragment in patients with previous joint surgery. *Eur J Nucl Med Mol Imaging*. 2002;29(4):547–551.
117. Stoeckli T.C, Zimmerli W, Maecke H.R, Fridrich R. Comparison of chemotaxis and superoxide generation of indium-111-oxine- and technetium-99m-HMPAO-labelled granulocytes. *Scand J Clin Lab Invest*. 1996;56(4):305–309.
118. Graute V, Feist M, Lehner S, et al. Detection of low-grade prosthetic joint infections using 99mTc-antigranulocyte SPECT/CT: initial clinical results. *Eur J Nucl Med Mol Imaging*. 2010;37(9):1751–1759.
119. Kiran M, Donnelly T.D, Armstrong C, Kapoor B, Kumar G, Peter V. Diagnostic utility of fluorodeoxyglucose positron emission tomography in prosthetic joint infection based on MSIS criteria. *Bone Joint J*. 2019;101-b(8):910–914.
120. Kumar R, Kumar R, Kumar V, Malhotra R. Potential clinical implication of (18) F-FDG PET/CT in diagnosis of periprosthetic infection and its comparison with (18) F-Fluoride PET/CT. *J Med Imaging Radiat Oncol*. 2016;60(3):315–322.
121. Corona P.S, Vicente M, Carrera L, Rodríguez-Pardo D, Corró S. Current actual success rate of the two-stage exchange arthroplasty strategy in chronic hip and knee periprosthetic joint infection. *Bone Joint J*. 2020;102-b(12):1682–1688.
- Byren I, Bejon P, Atkins B.L, et al. One hundred and twelve infected arthroplasties treated with 'DAIR' (debridement, antibiotics and implant retention): antibiotic duration and outcome. *J Antimicrob Chemother*. 2009;63(6):1264–1271.
- Puhto A.P, Puhto T, Syrjala H. Short-course antibiotics for prosthetic joint infections treated with prosthesis retention. *Clin Microbiol Infect*. 2012;18(11):1143–1148.
- Beswick A.D, Elvers K.T, Smith A.J, Gooberman-Hill R, Lovering A, Blom A.W. What is the evidence base to

- (debridement, antibiotics and implant retention), antibiotic duration and outcome. *J Antimicrob Chemother.* 2009;63(6):1264–1271.
- Puhto A.P, Puhto T, Syrjala H. Short-course antibiotics for prosthetic joint infections treated with prosthesis retention. *Clin Microbiol Infect.* 2012;18(11):1143–1148.
- Beswick A.D, Elvers K.T, Smith A.J, Gooberman-Hill R, Lovering A, Blom A.W. What is the evidence base to guide surgical treatment of infected hip prostheses? systematic review of longitudinal studies in unselected patients. *BMC Med.* 2012;10:18.
125. Bejon P, Berendt A, Atkins B.L, et al. Two-stage revision for prosthetic joint infection: predictors of outcome and the role of reimplantation microbiology. *J Antimicrob Chemother.* 2010;65(3):569–575.
126. El Helou O.C, Berbari E.F, Lahr B.D, et al. Efficacy and safety of rifampin containing regimen for staphylococcal prosthetic joint infections treated with debridement and retention. *Eur J Clin Microbiol Infect Dis.* 2010;29(8):961–967.
127. Zimmerli W, Ochsner P.E. Management of infection associated with prosthetic joints. *Infection.* 2003;31(2):99–108.
128. Achermann Y, Vogt M, Spormann C, et al. Characteristics and outcome of 27 elbow periprosthetic joint infections: results from a 14-year cohort study of 358 elbow prostheses. *Clin Microbiol Infect.* 2011;17(3):432–438.
129. Nelson G.N, Davis D.E, Namdari S. Outcomes in the treatment of periprosthetic joint infection after shoulder arthroplasty: a systematic review. *J Shoulder Elbow Surg.* 2016;25(8):1337–1345.
- Trebe R, Pisot V, Trampuz A. Treatment of infected retained implants. *J Bone Joint Surg Br.* 2005;87(2):249–256.
131. Sendi P, Zimmerli W. Challenges in periprosthetic knee-joint infection. *Int J Artif Organs.* 2011;34(9):947–956.
- Tschudin-Sutter S, Frei R, Dangel M, et al. Validation of a treatment algorithm for orthopaedic implant-related infections with device-retention-results from a prospective observational cohort study. *Clin Microbiol Infect.* 2016;22(5):457.e1–457.e9.
- Holmberg A, Thórhallsdóttir V.G, Robertsson O, A W.D, Stefánsdóttir A. 75% success rate after open debridement, exchange of tibial insert, and antibiotics in knee prosthetic joint infections. *Acta Orthop.* 2015;86(4):457–462.
134. Auñón Á, Tovar-Bazaga M, Blanco-García A, García-Cañete J, Parrón R, Esteban J. Does a new antibiotic scheme improve the outcome of *Staphylococcus aureus*-caused acute prosthetic joint infections (PJI) treated with debridement, antibiotics and implant retention (DAIR)? *Antibiotics (Basel).* 2022;11(7).
135. Jackson W.O, Schmalzried T.P. Limited role of direct exchange arthroplasty in the treatment of infected total hip replacements. *Clin Orthop Relat Res.* 2000;381:101–105.
136. Born P, Ilchmann T, Zimmerli W, et al. Eradication of infection, survival, and radiological results of uncemented revision stems in infected total hip arthroplasties. *Acta Orthop.* 2016;87(6):637–643.
137. Iarikov D, Demian H, Rubin D, Alexander J, Nambiar S. Choice and doses of antibacterial agents for cement spacers in treatment of prosthetic joint infections: review of published studies. *Clin Infect Dis.* 2012;55(11):1474–1480.
138. Winkler T, Stuhler M.G.W, Lieb E, et al. Outcome of short versus long interval in two-stage exchange for periprosthetic joint infection: a prospective cohort study. *Arch Orthop Trauma Surg.* 2019;139(3):295–303.
139. Sendi P, Rohrbach M, Graber P, Frei R, Ochsner P.E, Zimmerli W. *Staphylococcus aureus* small colony variants in prosthetic joint infection. *Clin Infect Dis.* 2006;43(8):961–967.
140. Bernard L, Arvieux C, Brunschweiler B, et al. Antibiotic therapy for 6 or 12 Weeks for prosthetic joint infection. *N Engl J Med.* 2021;384(21):1991–2001.
141. Li H K, Bombach I, Zambellas P, et al. Oral versus intravenous antibiotics for bone and joint infection. *N*

- Soriano A, García S, Bori G, et al. Treatment of acute post-surgical infection of joint arthroplasty. *Clin Microbiol Infect*. 2006;12(9):930–933.
143. Sendi P, Zimmerli W. The use of rifampin in staphylococcal orthopaedic-device-related infections. *Clin Microbiol Infect*. 2017;23(6):349–350.
144. Pushkin R, Iglesias-Ussel M.D, Keedy K, et al. A randomized study evaluating oral fusidic acid (CEM-102) in combination with oral rifampin compared with standard-of-care antibiotics for treatment of prosthetic joint infections: a newly identified drug-drug interaction. *Clin Infect Dis*. 2016;63(12):1599–1604.
145. Achermann Y, Eigenmann K, Ledergerber B, et al. Factors associated with rifampin resistance in staphylococcal periprosthetic joint infections (PJI): a matched case-control study. *Infection*. 2013;41(2):431–437.
146. van Hal S.J, Lodise T.P, Paterson D.L. The clinical significance of vancomycin minimum inhibitory concentration in *Staphylococcus aureus* infections: a systematic review and meta-analysis. *Clin Infect Dis*. 2012;54(6):755–771.
147. Perry C.M, Jarvis B. Linezolid: a review of its use in the management of serious gram-positive infections. *Drugs*. 2001;61(4):525–551.
148. Rao N, Hamilton C.W. Efficacy and safety of linezolid for gram-positive orthopedic infections: a prospective case series. *Diagn Microbiol Infect Dis*. 2007;59(2):173–179.
- Legout L, Valette M, Dezeque H, et al. Tolerability of prolonged linezolid therapy in bone and joint infection: protective effect of rifampicin on the occurrence of anaemia? *J Antimicrob Chemother*. 2010;65(10):2224–2230.
150. Razonable R.R, Osmon D.R, Steckelberg J.M. Linezolid therapy for orthopedic infections. *Mayo Clin Proc*. 2004;79(9):1137–1144.
151. Soriano A, Gómez J, Gómez L, et al. Efficacy and tolerability of prolonged linezolid therapy in the treatment of orthopedic implant infections. *Eur J Clin Microbiol Infect Dis*. 2007;26(5):353–356.
152. John A.K, Baldoni D, Haschke M, et al. Efficacy of daptomycin in implant-associated infection due to methicillin-resistant *Staphylococcus aureus*: importance of combination with rifampin. *Antimicrob Agents Chemother*. 2009;53(7):2719–2724.
153. Saleh-Mghir A, Muller-Serieys C, Dinh A, Massias L, Crémieux A.C. Adjunctive rifampin is crucial to optimizing daptomycin efficacy against rabbit prosthetic joint infection due to methicillin-resistant *Staphylococcus aureus*. *Antimicrob Agents Chemother*. 2011;55(10):4589–4593.
154. Lora-Tamayo J, Parra-Ruiz J, Rodríguez-Pardo D, et al. High doses of daptomycin (10 mg/kg/d) plus rifampin for the treatment of staphylococcal prosthetic joint infection managed with implant retention: a comparative study. *Diagn Microbiol Infect Dis*. 2014;80(1):66–71.
155. Kim P.W, Sorbello A.F, Wassel R.T, Pham T.M, Tønning J.M, Nambiar S. Eosinophilic pneumonia in patients treated with daptomycin: review of the literature and US FDA adverse event reporting system reports. *Drug Saf*. 2012;35(6):447–457.
156. Hsieh P.H, Lee M.S, Hsu K.Y, Chang Y.H, Shih H.N, Ueng S.W. Gram-negative prosthetic joint infections: risk factors and outcome of treatment. *Clin Infect Dis*. 2009;49(7):1036–1043.
- Zmistowski B, Fedorka C.J, Sheehan E, Deirmengian G, Austin M.S, Parvizi J. Prosthetic joint infection caused by gram-negative organisms. *J Arthroplasty*. 2011;26(6 Suppl):104–108.
158. Aboltins C.A, Dowsey M.M, Buisling K.L, et al. Gram-negative prosthetic joint infection treated with debridement, prosthesis retention and antibiotic regimens including a fluoroquinolone. *Clin Microbiol Infect*. 2011;17(6):862–867.

159. Marmarouk M, Marmarouk M, Marmarouk M, et al. Prosthetic joint infection: a systematic review of definition

158. Aboltins C.A, Dowsey M.M, Buising K.L, et al. Gram-negative prosthetic joint infection treated with debridement, prosthesis retention and antibiotic regimens including a fluoroquinolone. *Clin Microbiol Infect*. 2011;17(6):862–867.
159. Metsemakers W.J, Morgenstern M, McNally M.A, et al. Fracture-related infection: a consensus on definition from an international expert group. *Injury*. 2018;49(3):505–510.
160. Govaert G.A.M, Kuehl R, Atkins B.L, et al. Diagnosing fracture-related infection: current concepts and recommendations. *J Orthop Trauma*. 2020;34(1):8–17.
161. Onsea J, Van Lieshout E.M.M, Zalavras C, et al. Validation of the diagnostic criteria of the consensus definition of fracture-related infection. *Injury*. 2022;53(6):1867–1879.
162. Morgenstern M, Athanasou N.A, Ferguson J.Y, Metsemakers W.J, Atkins B.L, McNally M.A. The value of quantitative histology in the diagnosis of fracture-related infection. *Bone Joint J*. 2018;100-b(7):966–972.
163. Doshi P, Gopalan H, Sprague S, Pradhan C, Kulkarni S, Bhandari M. Incidence of infection following internal fixation of open and closed tibia fractures in India (INFINITI): a multi-centre observational cohort study. *BMC Musculoskelet Disord*. 2017;18(1):156.
164. Gustilo R.B, Merkow R.L, Templeman D. The management of open fractures. *J Bone Joint Surg Am*. 1990;72(2):299–304.
165. Papakostidis C, Kanakaris N.K, Pretel J, Faour O, Morell D.J, Giannoudis P.V. Prevalence of complications of open tibial shaft fractures stratified as per the Gustilo-Anderson classification. *Injury*. 2011;42(12):1408–1415.
166. Bhandari M, Petrisor B.A, Jeray K.J. Wound irrigation in initial management of open fractures. *N Engl J Med*. 2016;374(18):1789–1790.
167. Sendi P, Morgenstern M, Metsemakers W.-J, McNally M. Fracture-related infection of the long bones. In: Zimmerli W, ed. *Bone and Joint Infections: From Microbiology to Diagnostics and Treatment*. 2nd ed. Hoboken, NJ: John Wiley & Sons, Inc.; 2021:367–386.
168. Willenegger H, Roth B. [Treatment tactics and late results in early infection following osteosynthesis]. *Unfallchirurgie*. 1986;12(5):241–246 Behandlungstaktik und Spätergebnisse bei Frühinfekt nach Osteosynthese.
169. Depypere M, Sliepen J, Onsea J, et al. The microbiological etiology of fracture-related infection. *Front Cell Infect Microbiol*. 2022;12:934485.
170. Corrigan R.A, Sliepen J, Dudareva M, et al. Causative pathogens do not differ between early, delayed or late fracture-related infections. *Antibiotics (Basel)*. 2022;11(7):943.
171. Morgenstern M, Kuehl R, Zalavras C.G, et al. The influence of duration of infection on outcome of debridement and implant retention in fracture-related infection. *Bone Joint J*. 2021;103-b(2):213–221.
172. Libraty D.H, Patkar C, Torres B. *Staphylococcus aureus* reactivation osteomyelitis after 75 years. *N Engl J Med*. 2012;366(5):481–482.
173. Coiffier G, Ducasse E, Keraudren A, et al. Relevance of using both aerobic and anaerobic enrichment vials for optimizing rapid diagnosis of osteoarticular infections. *Microb Pathog*. 2022;165:105480.
174. Govaert G.A, FF I.J, McNally M, McNally E, Reininga I.H, Glaudemans A.W. Accuracy of diagnostic imaging modalities for peripheral post-traumatic osteomyelitis - a systematic review of the recent literature. *Eur J Nucl Med Mol Imaging*. 2017;44(8):1393–1407.
175. Govaert G.A.M, Glaudemans A. Nuclear medicine imaging of posttraumatic osteomyelitis. *Eur J Trauma Emerg Surg*. 2016;42(4):397–410.
176. Worlock P, Slack R, Harvey L, Mawhinney R. The prevention of infection in open fractures: an experimental study of the effect of fracture stability. *Injury*. 1994;25(1):31–38.

- literature. *Eur J Nucl Med Mol Imaging*. 2017;44(8):1393–1407.
175. Govaert G.A.M, Glaudemans A. Nuclear medicine imaging of posttraumatic osteomyelitis. *Eur J Trauma Emerg Surg*. 2016;42(4):397–410.
 176. Worlock P, Slack R, Harvey L, Mawhinney R. The prevention of infection in open fractures: an experimental study of the effect of fracture stability. *Injury*. 1994;25(1):31–38.
 177. Schmidt A.H, Swiontkowski M.F. Pathophysiology of infections after internal fixation of fractures. *J Am Acad Orthop Surg*. 2000;8(5):285–291.
 178. Zalavras C.G. Prevention of infection in open fractures. *Infect Dis Clin North Am*. 2017;31(2):339–352.
 179. Rupp M, Popp D, Alt V. Prevention of infection in open fractures: where are the pendulums now? *Injury*. 2020;51(Suppl 2):S57–s63.
 180. Athans V, Veve M.P, Davis S.L. Trowels and Tribulations: review of antimicrobial-impregnated bone cements in prosthetic joint surgery. *Pharmacotherapy*. 2017;37(12):1565–1577.
 181. Morgenstern M, Vallejo A, McNally M.A, et al. The effect of local antibiotic prophylaxis when treating open limb fractures. *Bone Joint Res*. 2018;7(7):447–456.
 182. Thomes B, Murray P, Bouchier-Hayes D. Development of resistant strains of *Staphylococcus epidermidis* on gentamicin-loaded bone cement in vivo. *J Bone Joint Surg Br*. 2002;84(5):758–760.
 183. Neut D, van de Belt H, Stokroos I, van Horn J.R, van der Mei H.C, Busscher H.J. Biomaterial-associated infection of gentamicin-loaded PMMA beads in orthopaedic revision surgery. *J Antimicrob Chemother*. 2001;47(6):885–891.
 184. Menge T.J, Koethe J.R, Jenkins C.A, et al. Acute kidney injury after placement of an antibiotic-impregnated cement spacer during revision total knee arthroplasty. *J Arthroplasty*. 2012;27(6):1221–1227.e1–2.
 185. McNally M.A, Ferguson J.Y, Scarborough M, Ramsden A, Stubbs D.A, Atkins B.L. Mid- to long-term results of single-stage surgery for patients with chronic osteomyelitis using a bioabsorbable gentamicin-loaded ceramic carrier. *Bone Joint J*. 2022;104-b(9):1095–1100.
 186. Lethaby A, Temple J, Santy-Tomlinson J. Pin site care for preventing infections associated with external bone fixators and pins. *Cochrane Database Syst Rev*. 2013(12):Cd004551.
 187. Iobst C, Liu R. A systematic review of incidence of pin track infections associated with external fixation. Review Article. *J Limb Lengthening Reconstr*. 2016;2(1):6–16.
 188. Hill C, Flamant R, Mazas F, Evrard J. Prophylactic cefazolin versus placebo in total hip replacement. Report of a multicentre double-blind randomised trial. *Lancet*. 1981;1(8224):795–796.
 189. Bryson D.J, Morris D.L, Shivji F.S, Rollins K.R, Snape S, Ollivere B.J. Antibiotic prophylaxis in orthopaedic surgery: difficult decisions in an era of evolving antibiotic resistance. *Bone Joint J*. 2016;98-b(8):1014–1019.
 190. Bratzler D.W, Houck P.M. Antimicrobial prophylaxis for surgery: an advisory statement from the National Surgical Infection Prevention Project. *Clin Infect Dis*. 2004;38(12):1706–1715.
 191. van Kasteren M.E, Manniën J, Ott A, Kullberg B.J, de Boer A.S, Gyssens I.C. Antibiotic prophylaxis and the risk of surgical site infections following total hip arthroplasty: timely administration is the most important factor. *Clin Infect Dis*. 2007;44(7):921–927.
 192. Steinberg J.P, Braun B.I, Hellinger W.C, et al. Timing of antimicrobial prophylaxis and the risk of surgical site infections: results from the Trial to Reduce Antimicrobial Prophylaxis Errors. *Ann Surg*. 2009;250(1):10–16.
 193. Hawn M.T, Richman J.S, Vick C.C, et al. Timing of surgical antibiotic prophylaxis and the risk of surgical site infection. *JAMA Surg*. 2013;148(7):649–657.
 194. Weber W.P, Mujagic E, Zwahlen M, et al. Timing of surgical antimicrobial prophylaxis: a phase 3 randomised controlled trial. *Lancet Infect Dis*. 2017;17(6):605–614.

- **212.** Berbari E.F, Osmon D.R, Carr A, et al. Dental procedures as risk factors for prosthetic hip or knee infection: a hospital-based prospective case-control study. *Clin Infect Dis.* 2010;50(1):8–16.
- **213.** Thornhill M.H, Crum A, Rex S, et al. Analysis of prosthetic joint infections following invasive dental procedures in England. *JAMA Netw Open.* 2022;5(1):e2142987.
- **214.** Zimmerli W, Sendi P. Antibiotics for prevention of periprosthetic joint infection following dentistry: time to focus on data. *Clin Infect Dis.* 2010;50(1):17–19.
- **215.** Gupta A, Osmon D.R, Hanssen A.D, et al. Genitourinary procedures as risk factors for prosthetic hip or knee infection: a hospital-based prospective case-control study. *Open Forum Infect Dis.* 2015;2(3):ofv097.
- **216.** Rybak M.J, Lomaestro B.M, Rotschafer J.C, et al. Vancomycin therapeutic guidelines: a summary of consensus recommendations from the infectious diseases society of America, the American society of health-system pharmacists, and the society of infectious diseases pharmacists. *Clin Infect Dis.* 2009;49(3):325–327.
- **217.** El Helou O.C, Berbari E.F, Marculescu C.E, et al. Outcome of enterococcal prosthetic joint infection: is combination systemic therapy superior to monotherapy? *Clin Infect Dis.* 2008;47(7):903–909.

^a References 3, 33, 34, 53, 54, 64, 122–125.

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- **212.** Berbari E.F, Osmon D.R, Carr A, et al. Dental procedures as risk factors for prosthetic hip or knee infection: a hospital-based prospective case-control study. *Clin Infect Dis.* 2010;50(1):8–16.
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Anogenital Skin and Mucous Membrane Lesions

Michael H. Augenbraun

Short View Summary

Definition

- Infectious anogenital skin and mucous membrane lesions cover a wide spectrum of pathologic etiologies and clinical presentations, mostly manifesting as inflammatory reactions or defects in the epithelium of the genital tract in men and women.

Epidemiology

- Infectious anogenital skin and mucous membrane lesions are seen globally without seasonality.
- These lesions occur primarily in sexually active individuals.
- Most are caused by sexually transmitted infections, but neoplastic and noninfectious inflammatory processes should be considered, especially in older individuals.
- Incubation periods may vary according to etiology: for human papillomavirus (HPV), it may potentially be weeks, months, or years after exposure; for primary syphilis, it may be weeks to months; for herpes simplex virus (HSV) outbreaks, it may be years.
- Disease may be modified in populations that are immunocompromised.
- Some conditions are currently being seen more frequently in men who have sex with men (i.e., early-stage syphilis, lymphogranuloma venereum [LGV]).

Microbiology

- Bacteria: *Treponema pallidum* (spirochete), *Haemophilus ducreyi* (gram-negative diplococcus), *Chlamydia trachomatis* L serovars (obligate intracellular pathogen), *Klebsiella granulomatis*/donovanosis (gram-negative rod)
- Viruses: HSV, molluscum contagiosum virus, HPV, mpox virus
- Fungi: *Candida albicans*

Diagnosis

- Clinical assessment alone may be misleading.
- Laboratory examination using microscopy, nucleic acid amplification tests (NAATs), and serologic testing is often helpful.
- Gram staining and light microscopy help to identify *H. ducreyi* and *Candida* spp.
- Darkfield microscopy is used to identify *T. pallidum*.
- Serology helps to diagnose *T. pallidum* infection.
- NAATs and, less often, cell culture testing are used to diagnose HSV.
- NAATs are used to identify *C. trachomatis*.

Therapy

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